Thesis progress

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UNIVERSITÀ
DEGLI STUDI
DI MILANO

Index

1 1° week

2 2° week

3 3° week

GitHub

The whole code is available at this GitHib link

1 ° week

Uploading file .root

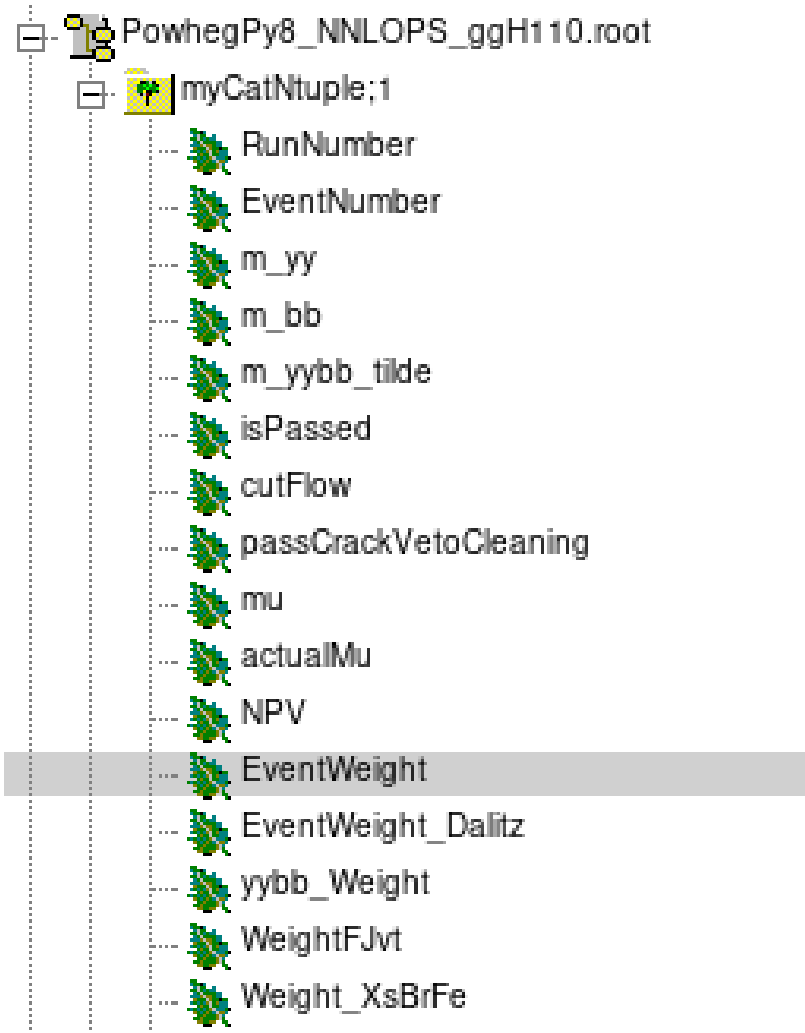
The file uploaded are:

- PowhegPy8_NNLOPS_ggH110.root;
- PowhegPy8_NNLOPS_ggH125.root;
- PowhegPy8_NNLOPS_ggH130.root;
- PowhegPy8_NNLOPS_ggH140.root;

Data

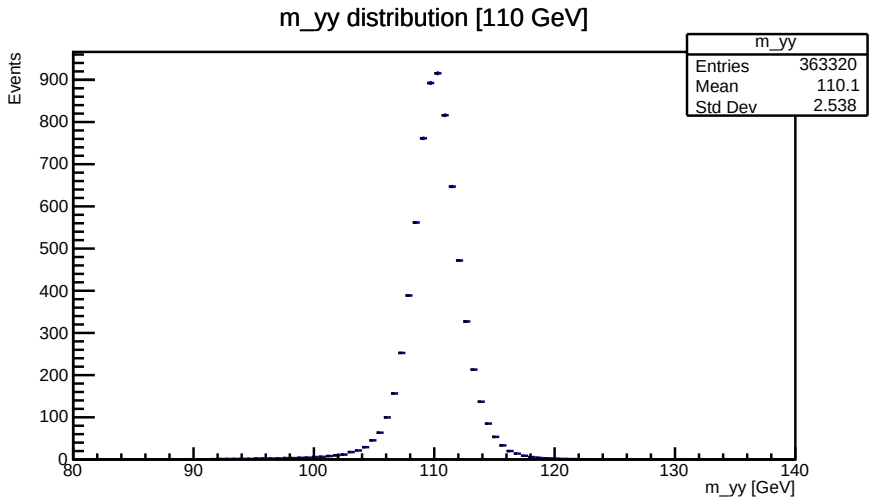
In each file, there is a `TTree myCatNtuple`, which contains all the variables. Only three of them were used:

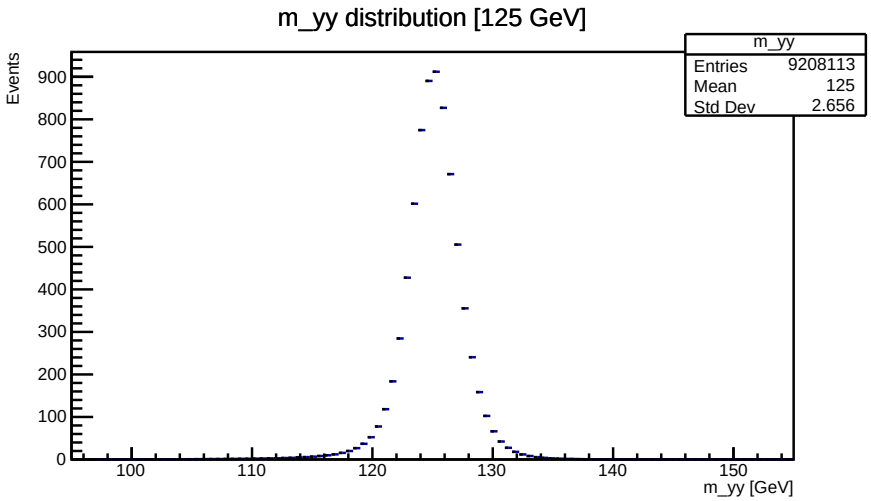
- **m_yy**: mass distribution;
- **EventWeight**: it contains all type of weights.
- **cutFlow**

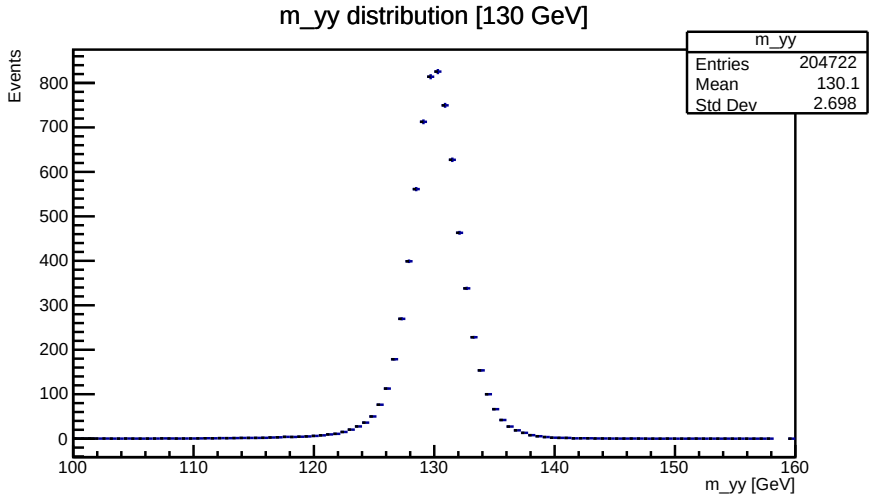


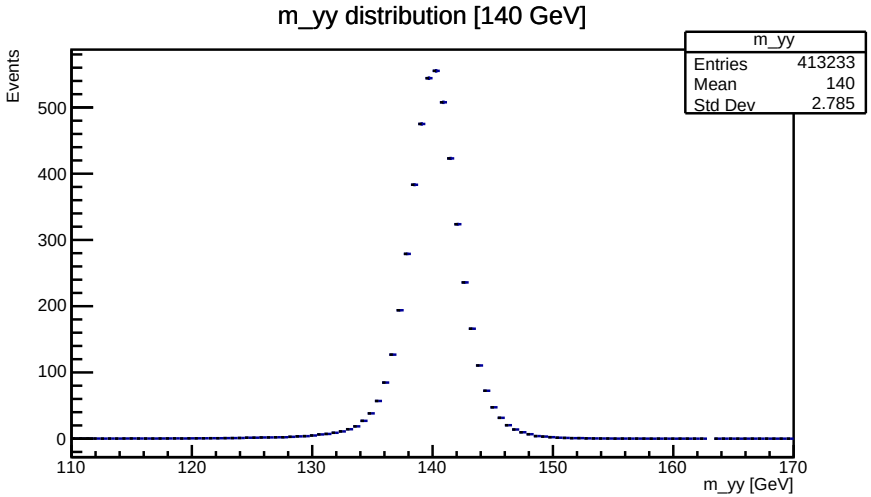
$m_{\gamma\gamma}$ distibution

Using all entries the distributions at different masses are:









Data analysis and fits

The distributions are fitted using two type of fit:

- **gaussian fit**: there are 2 parameters:
  - mean μ;
  - sigma σ;

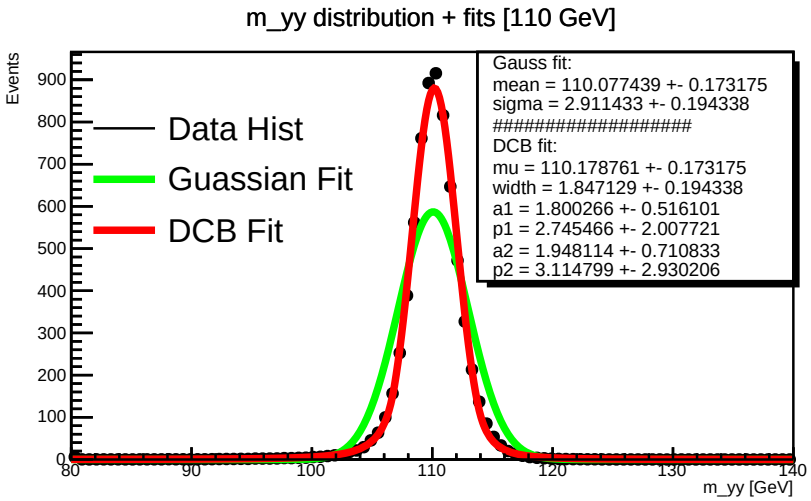
$$f(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

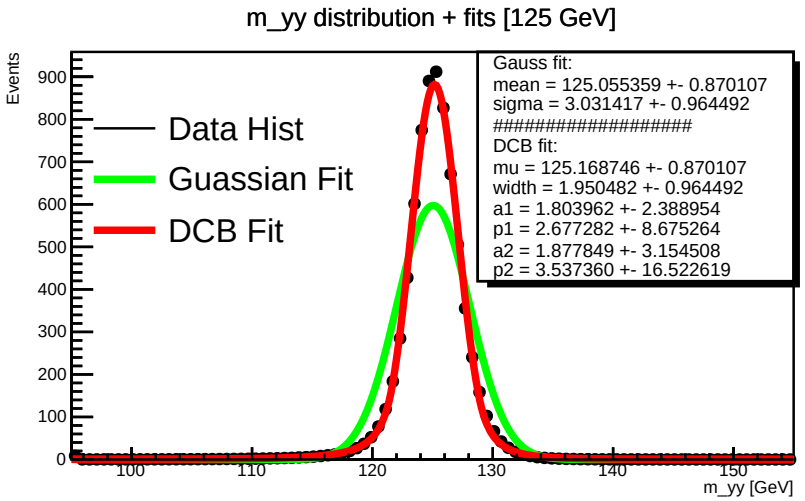
- **DCB fit**: there are six parameters:
  - mean μ;
  - sigma σ
  - a_1;
  - p_1;
  - a_2;
  - p_2;

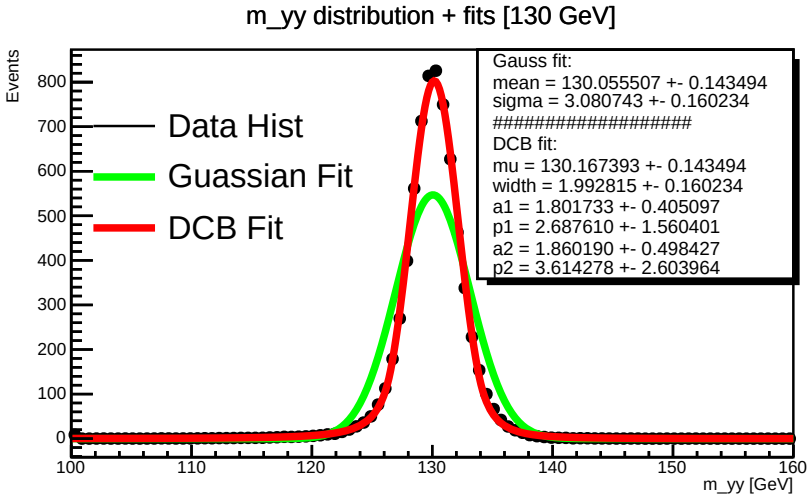
$$f(x, \mu, \sigma, a_1, p_1, a_2, p_2) = \cdots$$

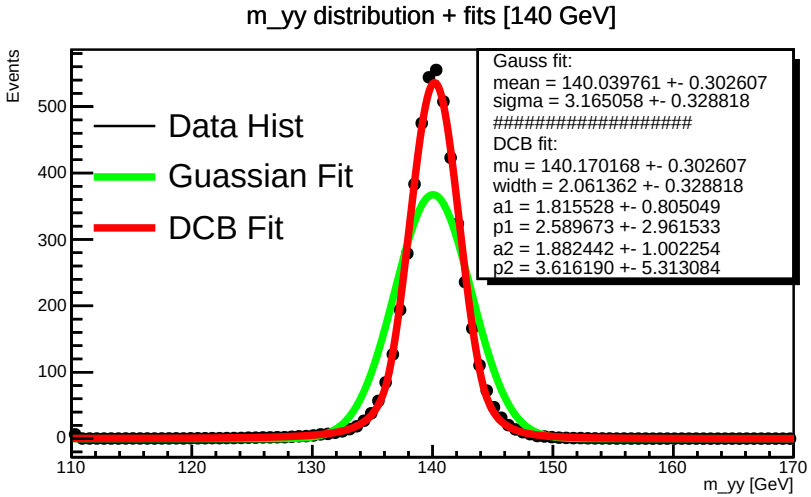
Gaussian and DCB fits

Applying the fits to all entries:









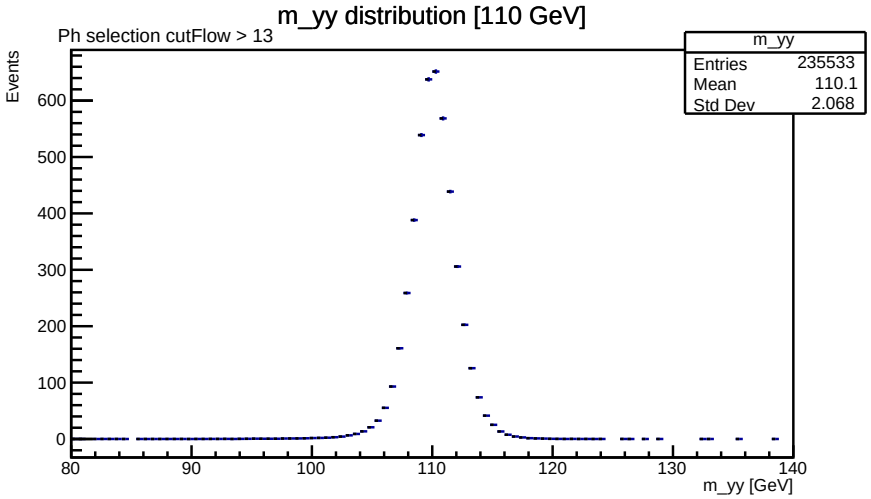
CutFlow

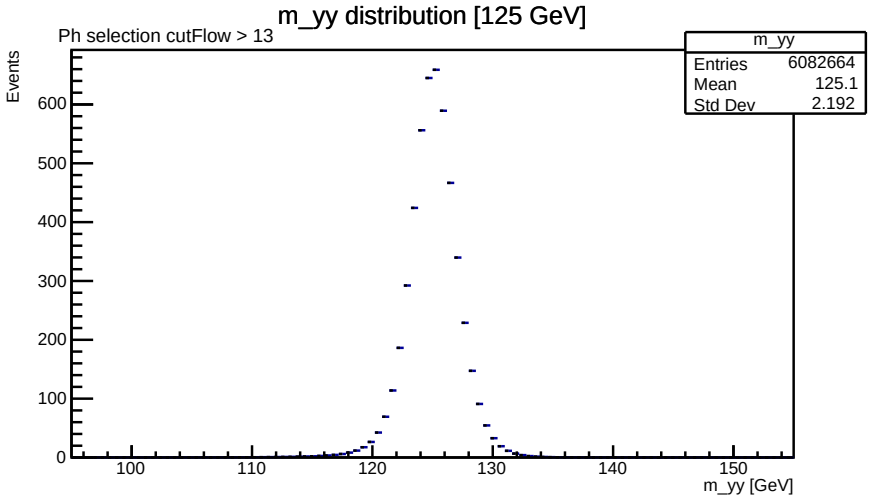
Events with at least two isolated γ with tight identification could be selected, where the transverse momentum of the leading photon is such that $p_T/m_{\gamma\gamma} > 0.35$, whereas that of the sub-leading photon is $p_T/m_{\gamma\gamma} > 0.25$. Then there are also other demands on the "quality" of data collection and triggers.

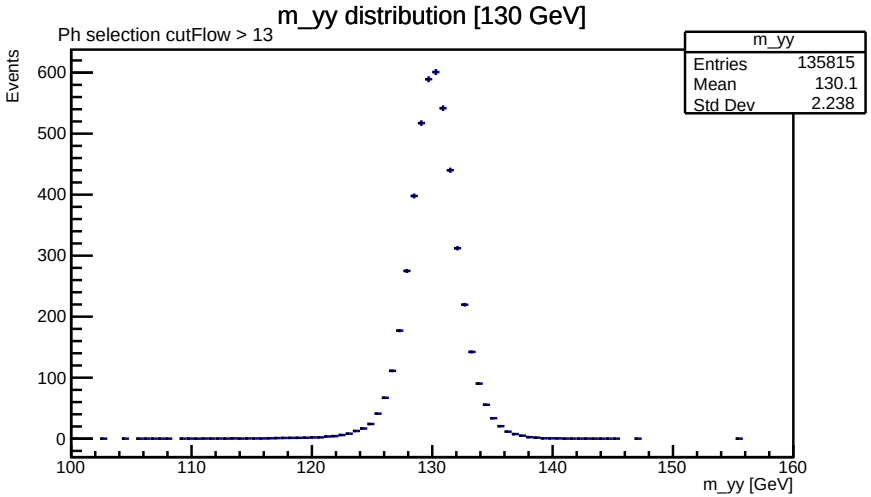
$\Longrightarrow$ Using the `cutFlow` greater than 13, the events are selected.

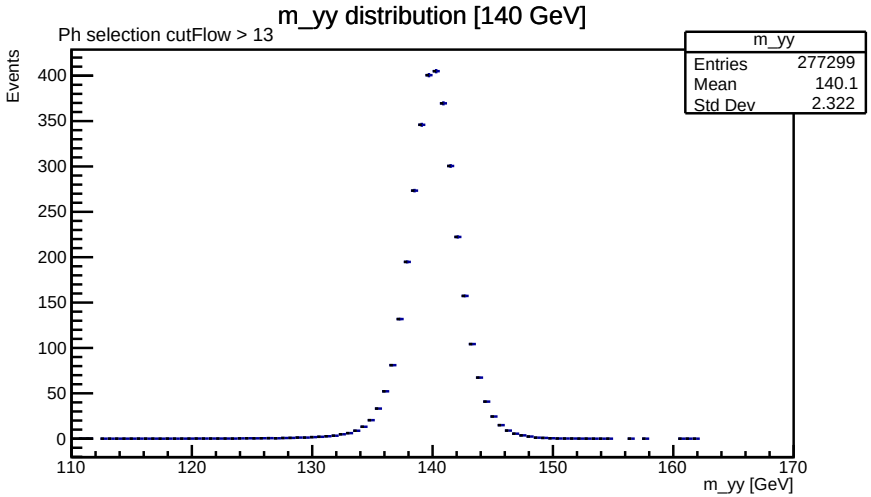
$m_{\gamma\gamma}$ distribution with cutFlow > 13

Using `cutFlow`>13 entries the distributions at different masses are:



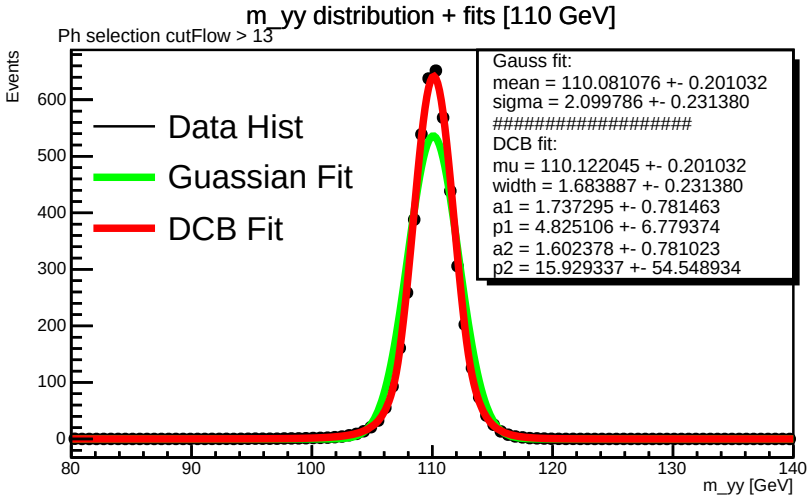


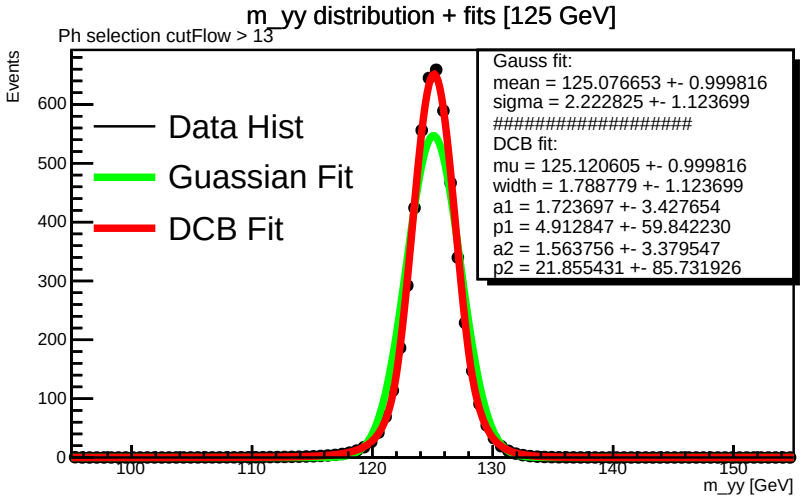


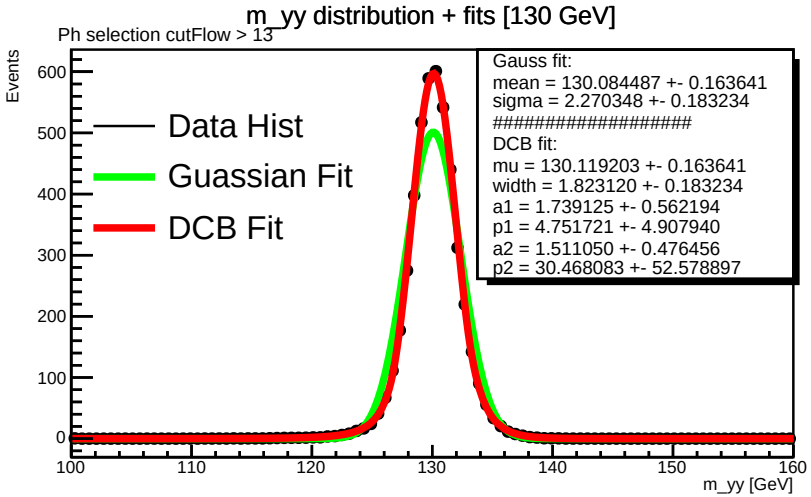


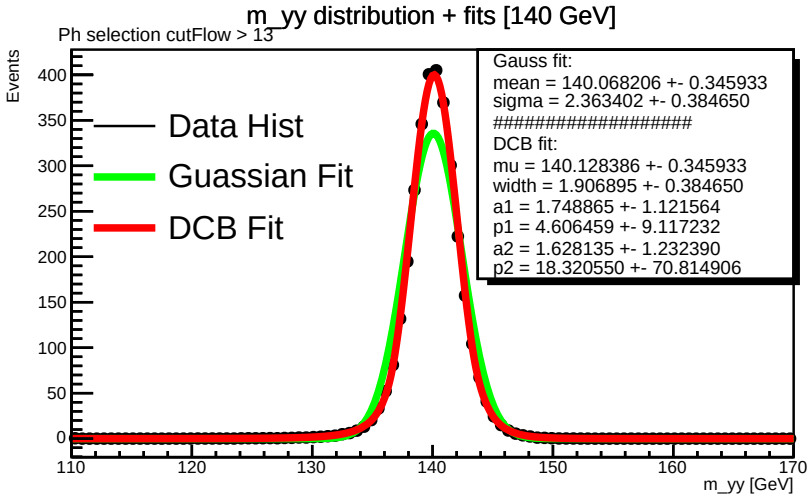
Gaussian and DCB fits with cutFlow > 13

Applying the fits to all entries:



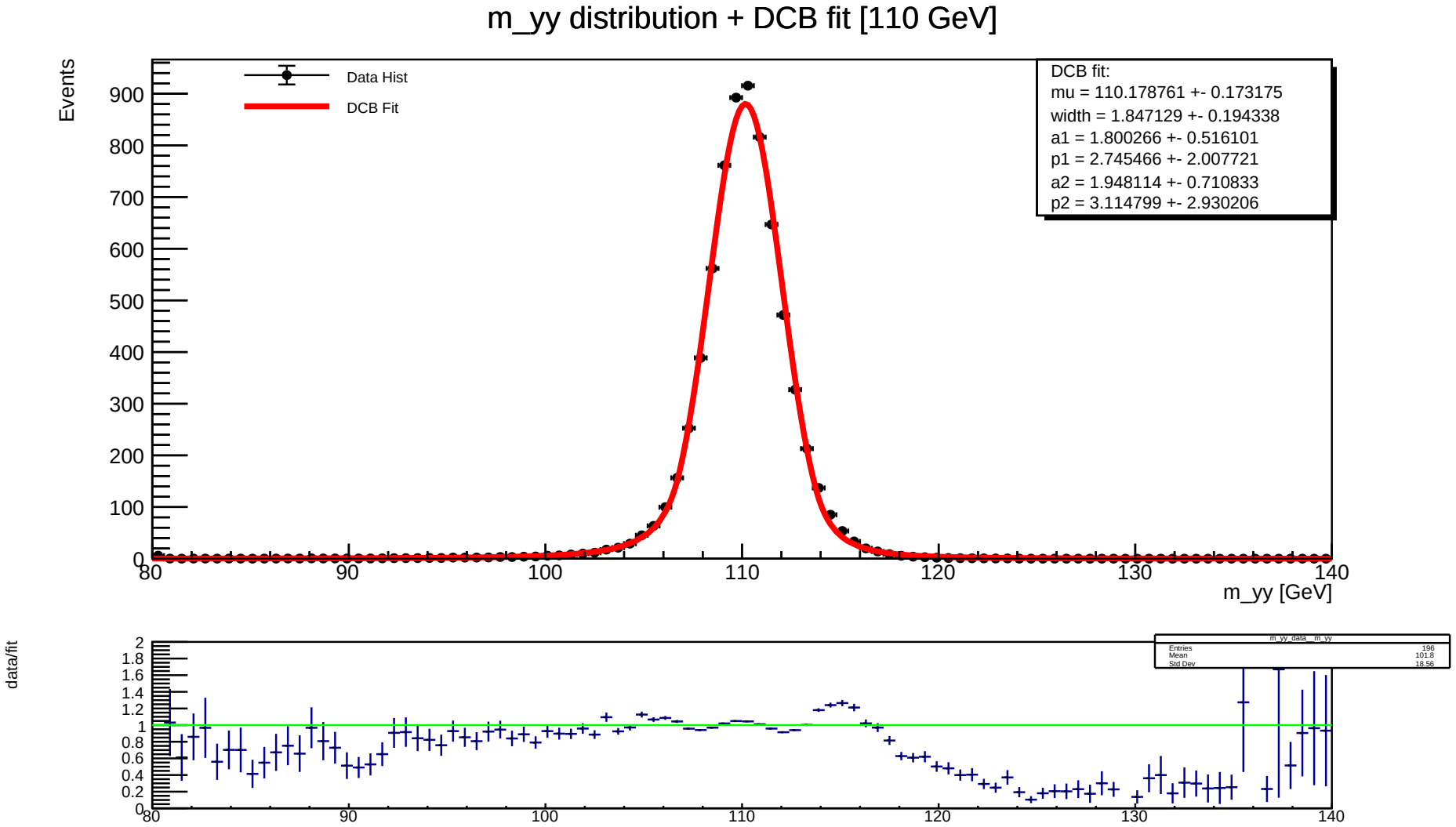






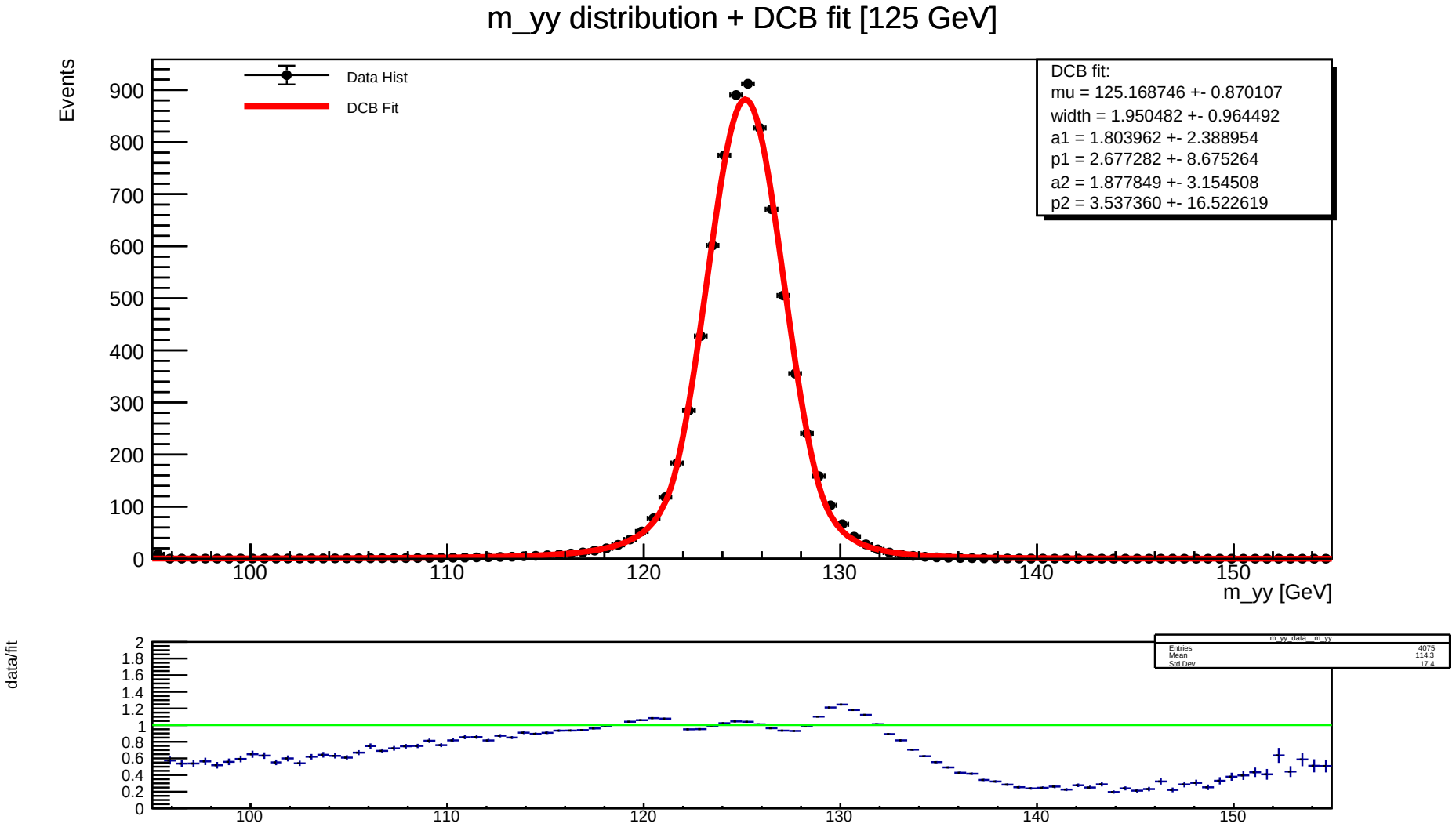
DCB fit and ratio plot [110 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 110 GeV mass distribution:



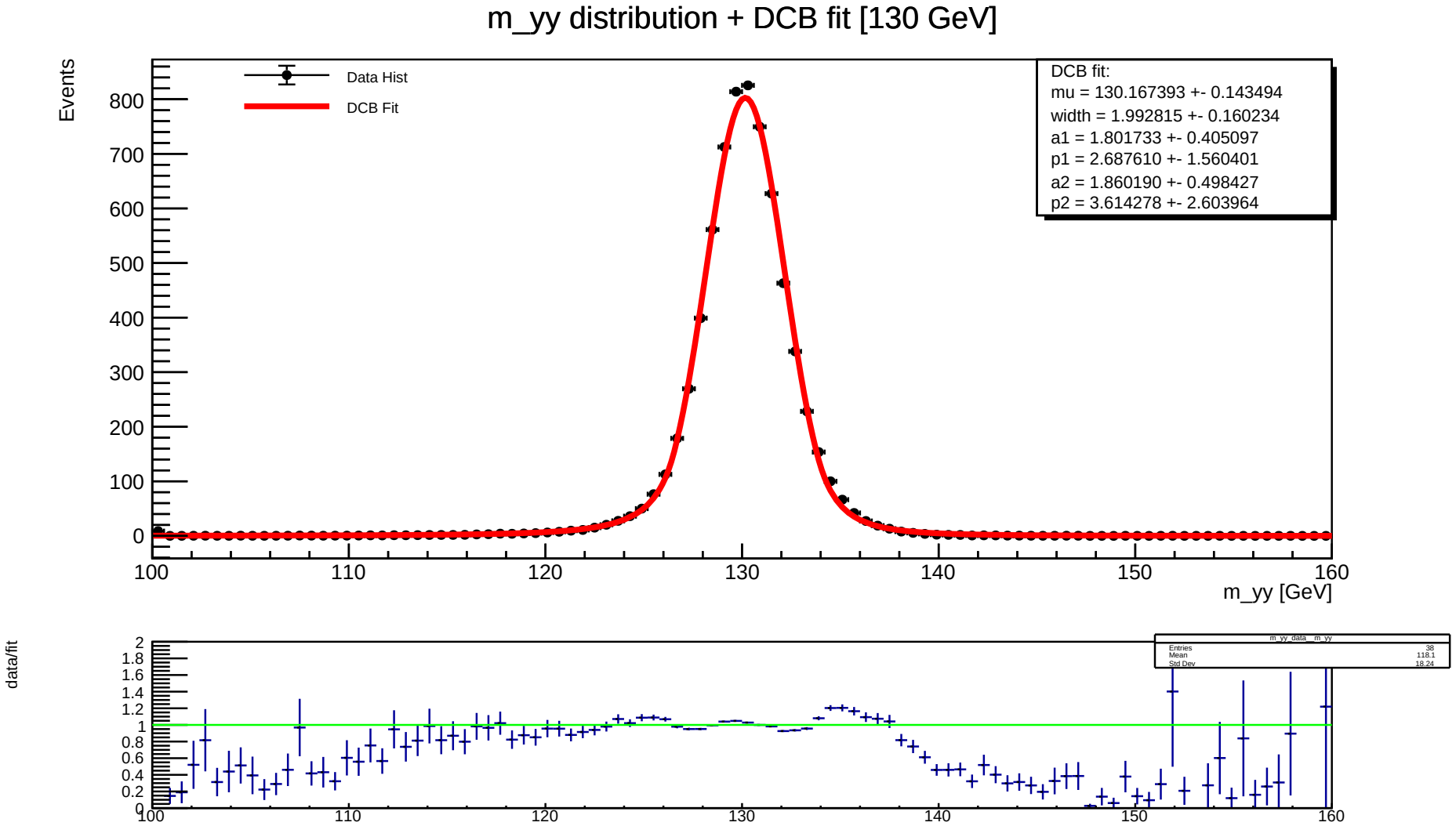
DCB fit and ratio plot [125 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 125 GeV mass distribution:



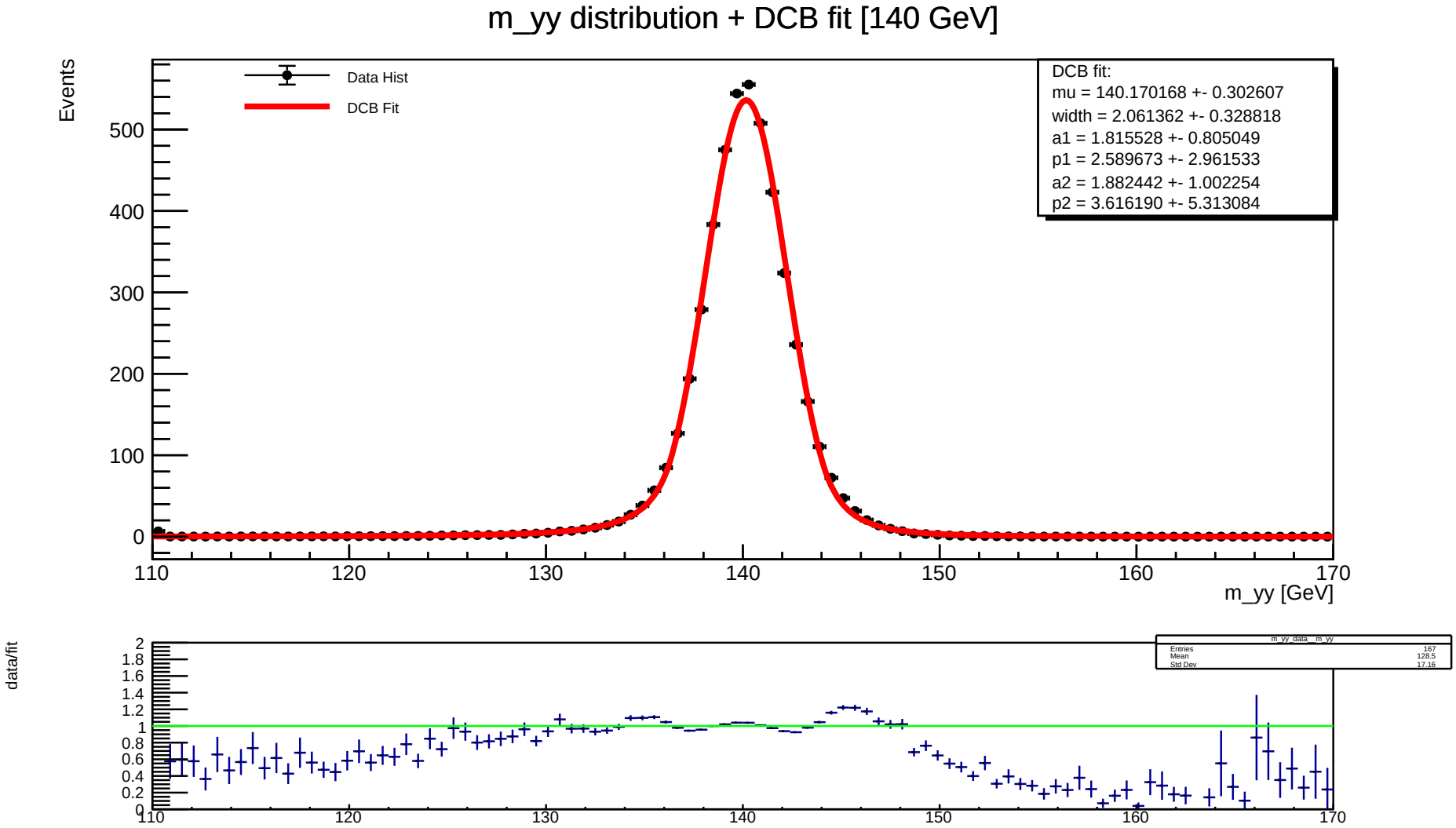
DCB fit and ratio plot [130 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 130 GeV mass distribution:



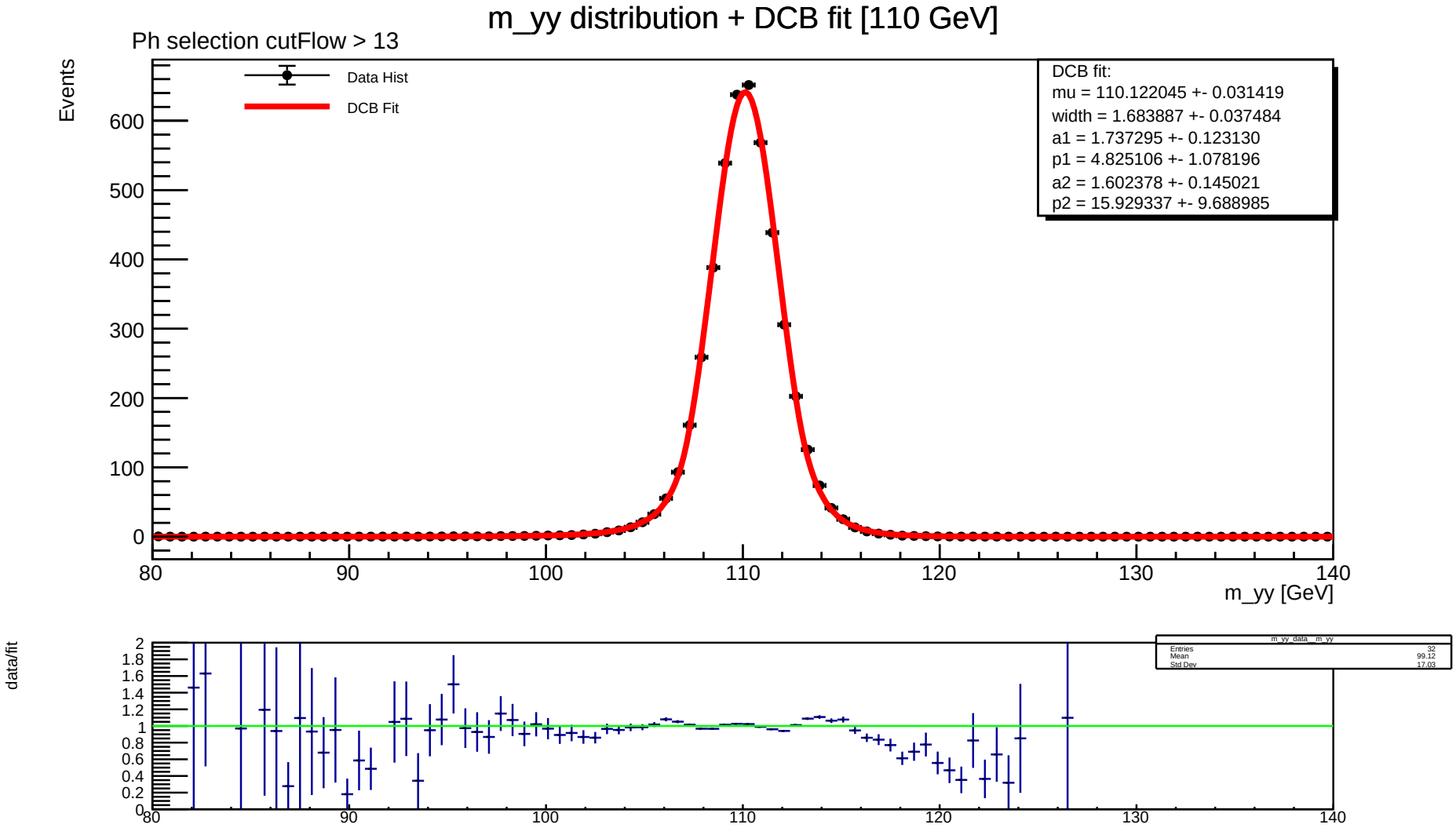
DCB fit and ratio plot [140 GeV]

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 140 GeV mass distribution:



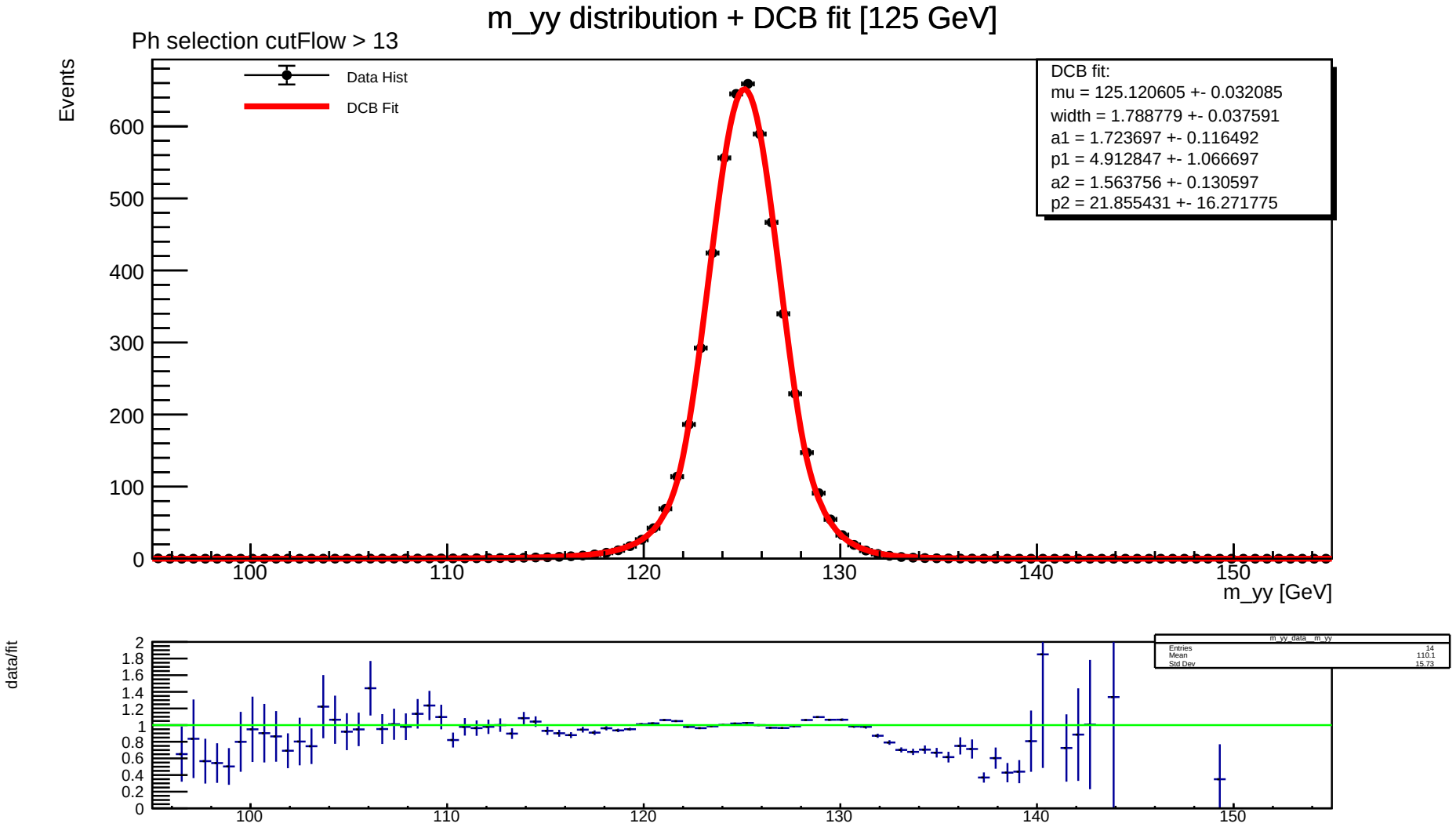
DCB fit+ratioplot [110 GeV] (cutFlow > 13)

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 110 GeV mass distribution with `cutFlow` > 13:



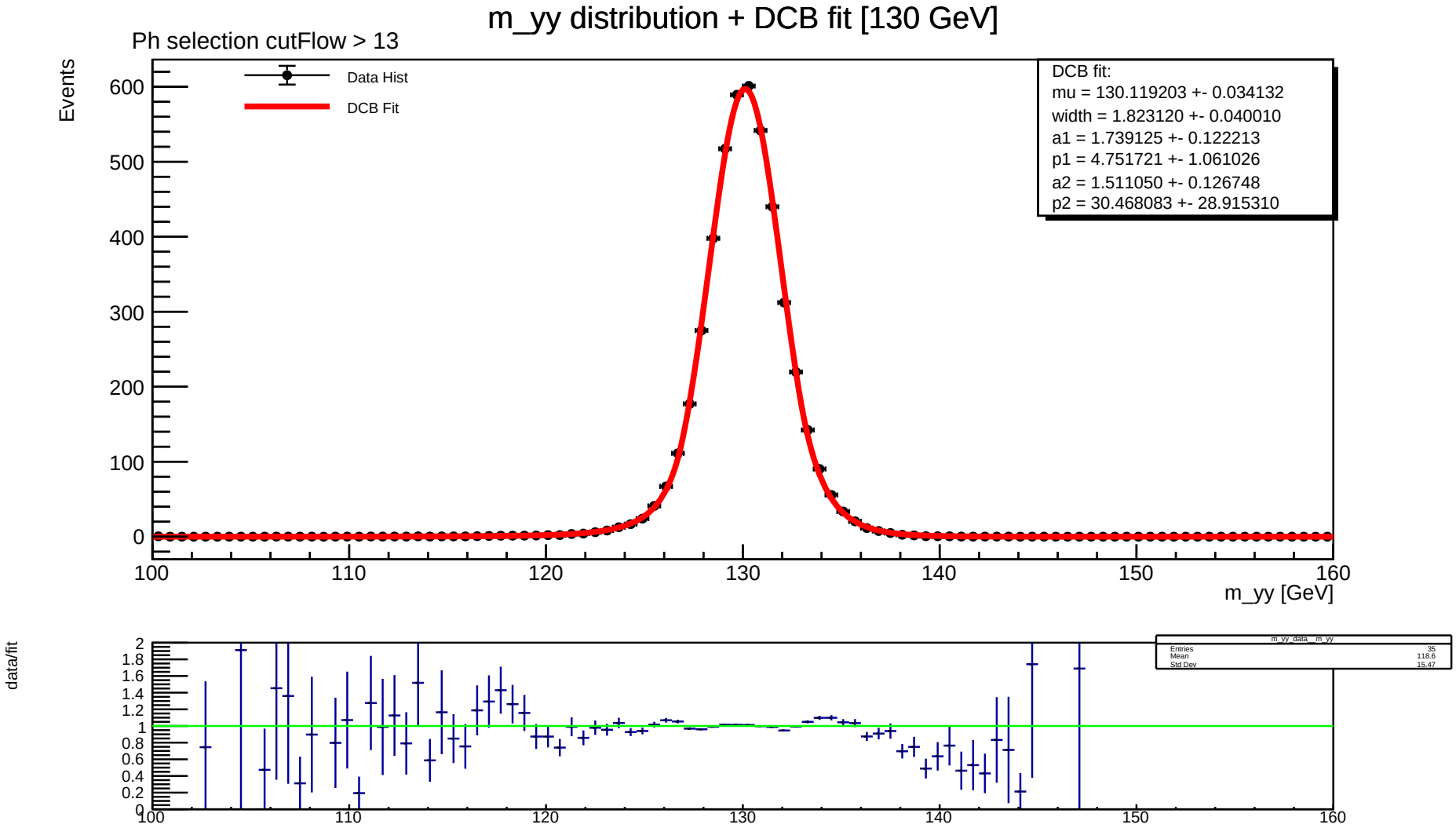
DCB fit+ratioplot [125 GeV] (cutFlow > 13)

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 125 GeV mass distribution with `cutFlow` > 13:



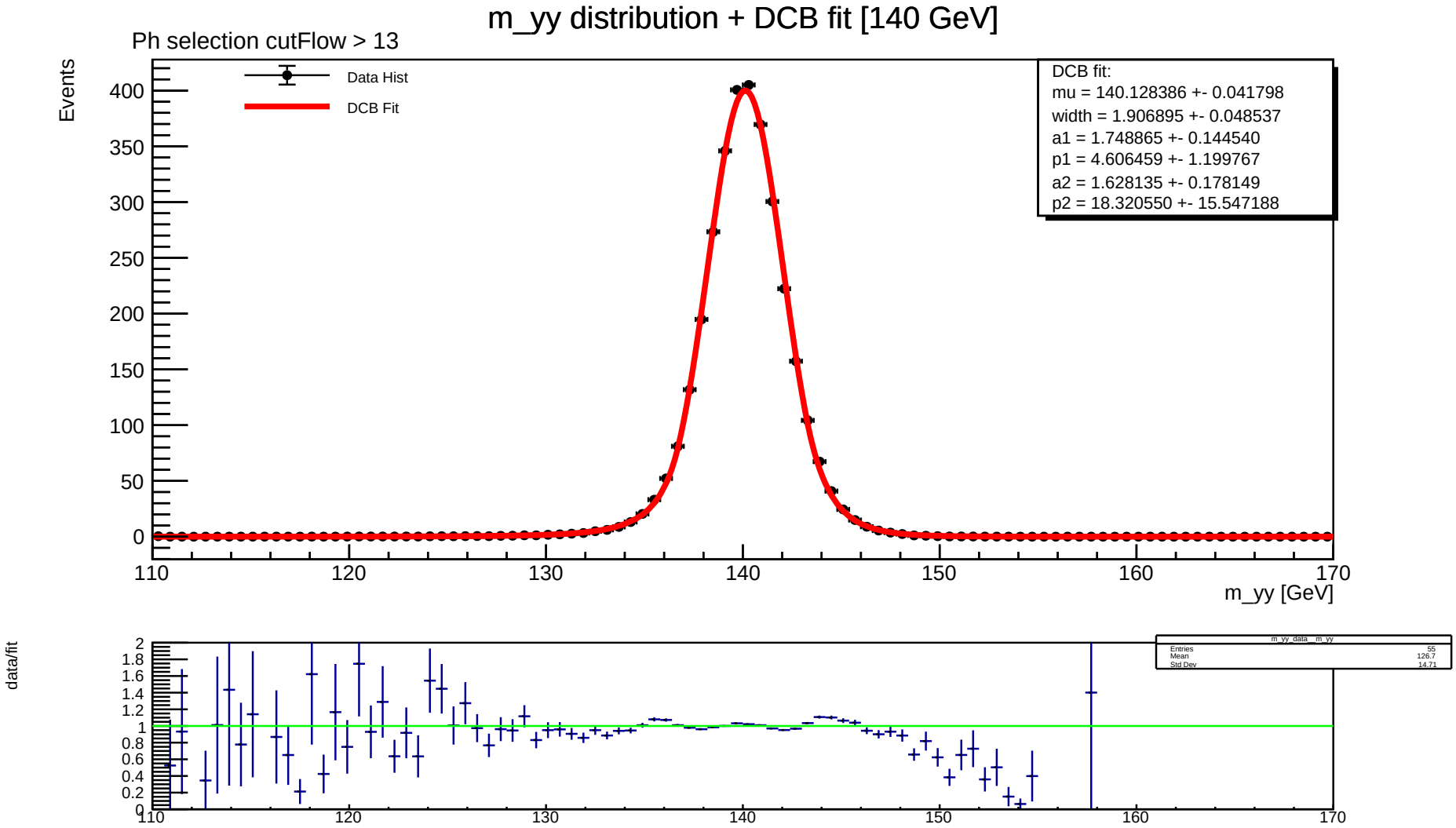
DCB fit+ratioplot [130 GeV] (cutFlow > 13)

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 130 GeV mass distribution with `cutFlow` > 13:



DCB fit+ratioplot [140 GeV] (cutFlow > 13)

The `ratio plot` between the fit and the distribution is added to each DCB fit. Using the 140 GeV mass distribution with `cutFlow` > 13:



2° week

DSCB(m_h)

The DSCD fit are studied with 3 different methods, in addiction to all free parameters fit:

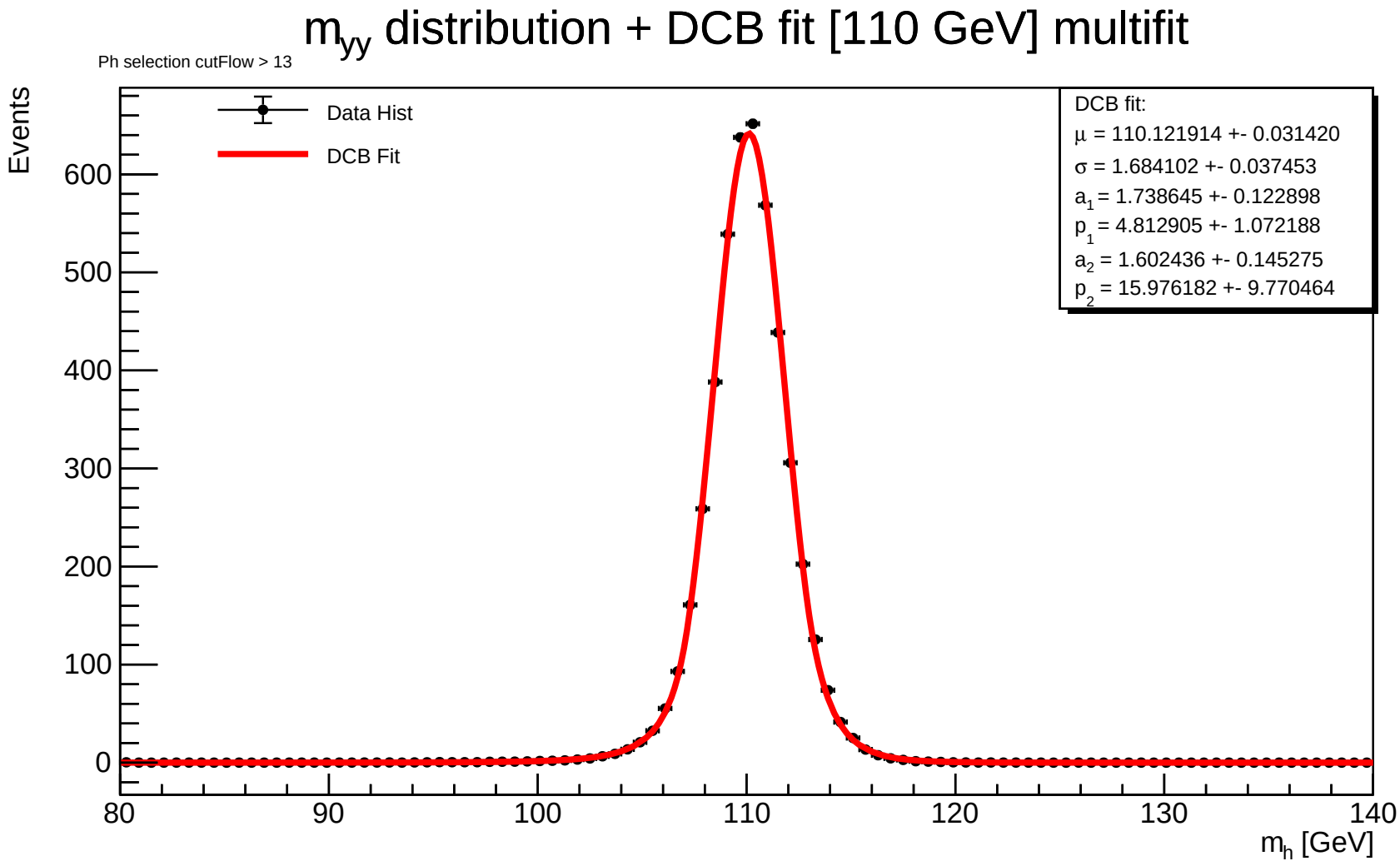
1. multi-fit;
2. μ and σ are functions of m_h;
3. all parameters are functions of m_h;

DSCB multifit

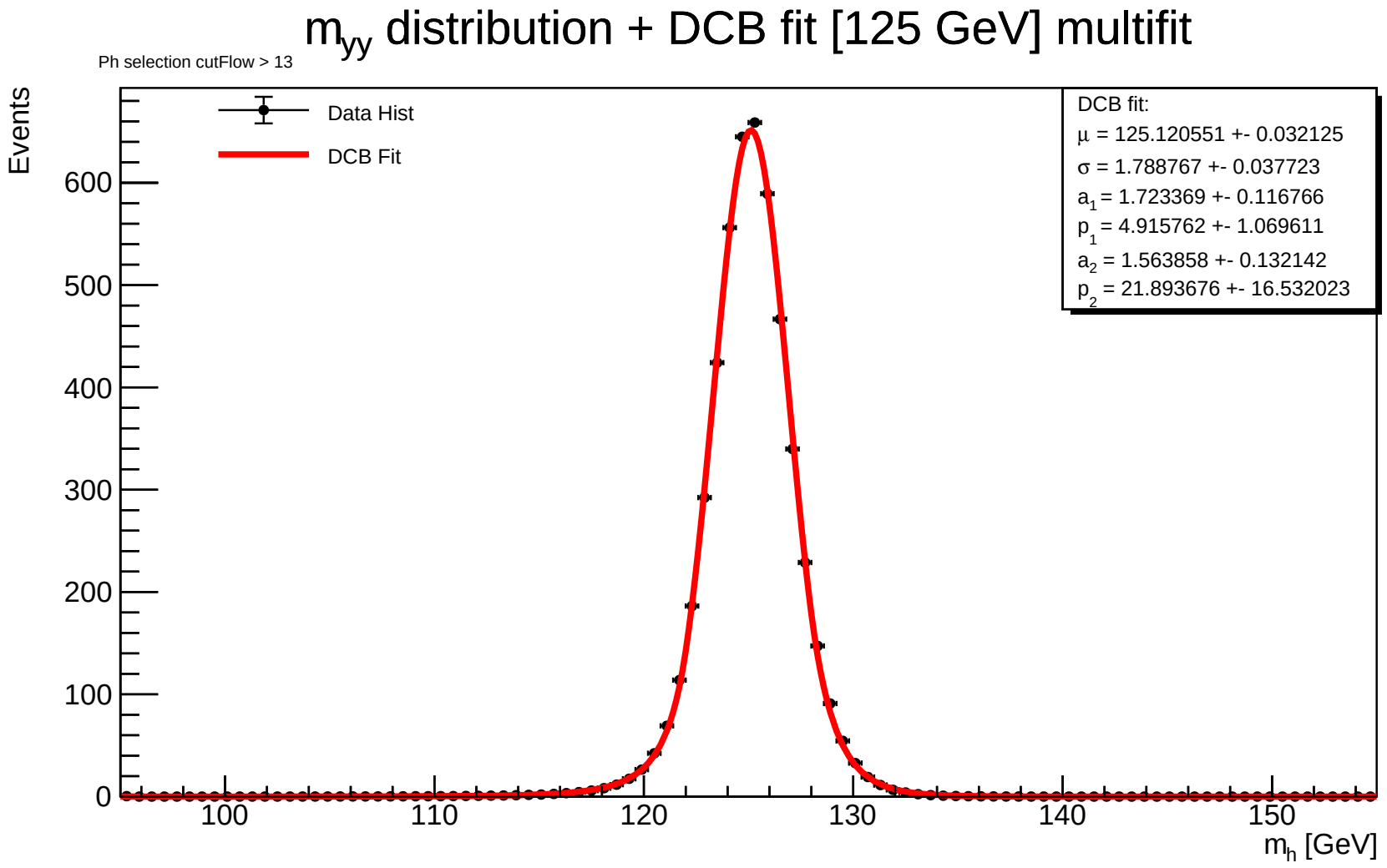
A DSCB multifit consists of a series of consecutive fits:

1. fit on μ and σ, tails params fixed;
2. fit on tails params, μ and σ fixed;
3. fit on all parameters;

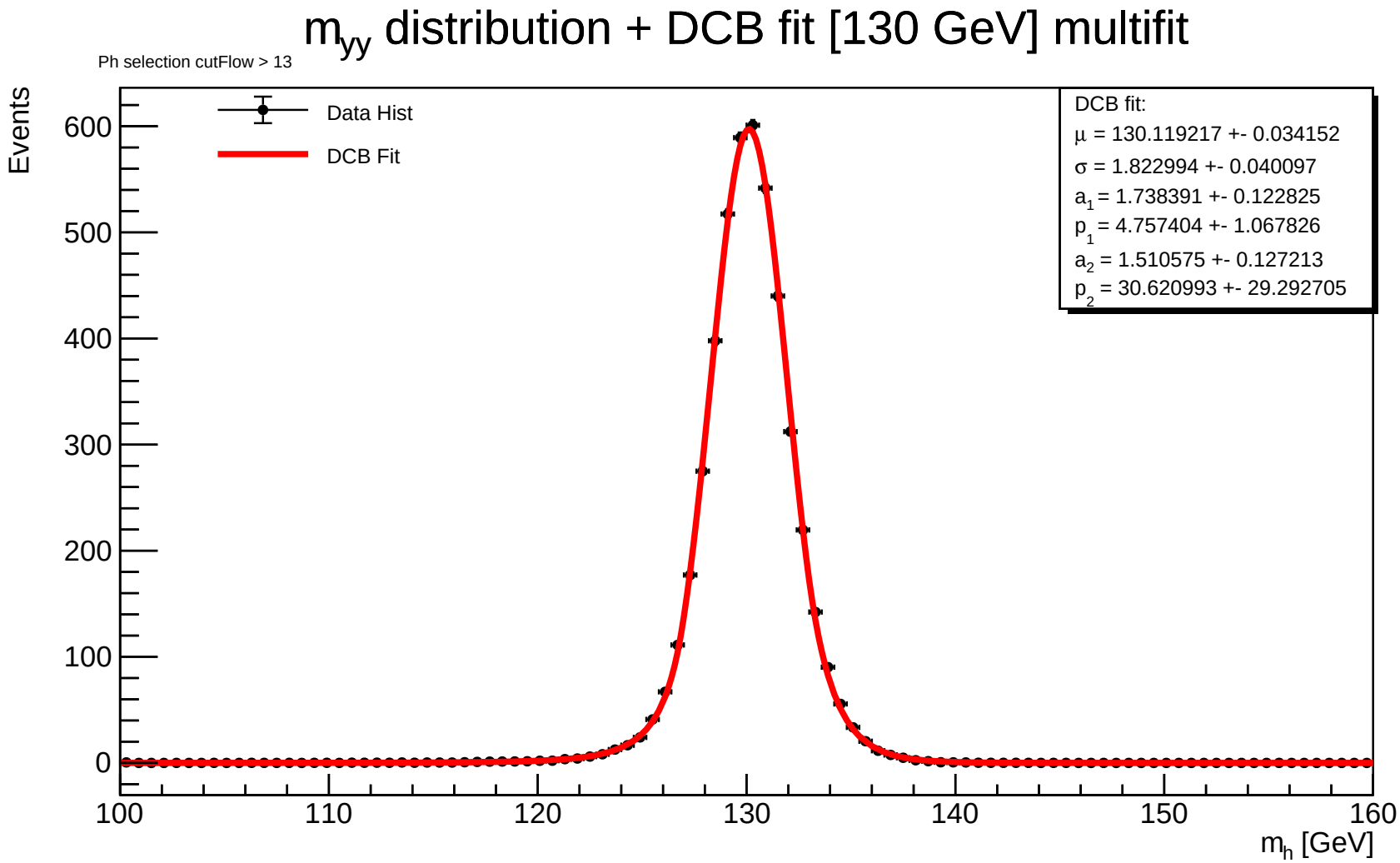
DSCB multifit [110 GeV]



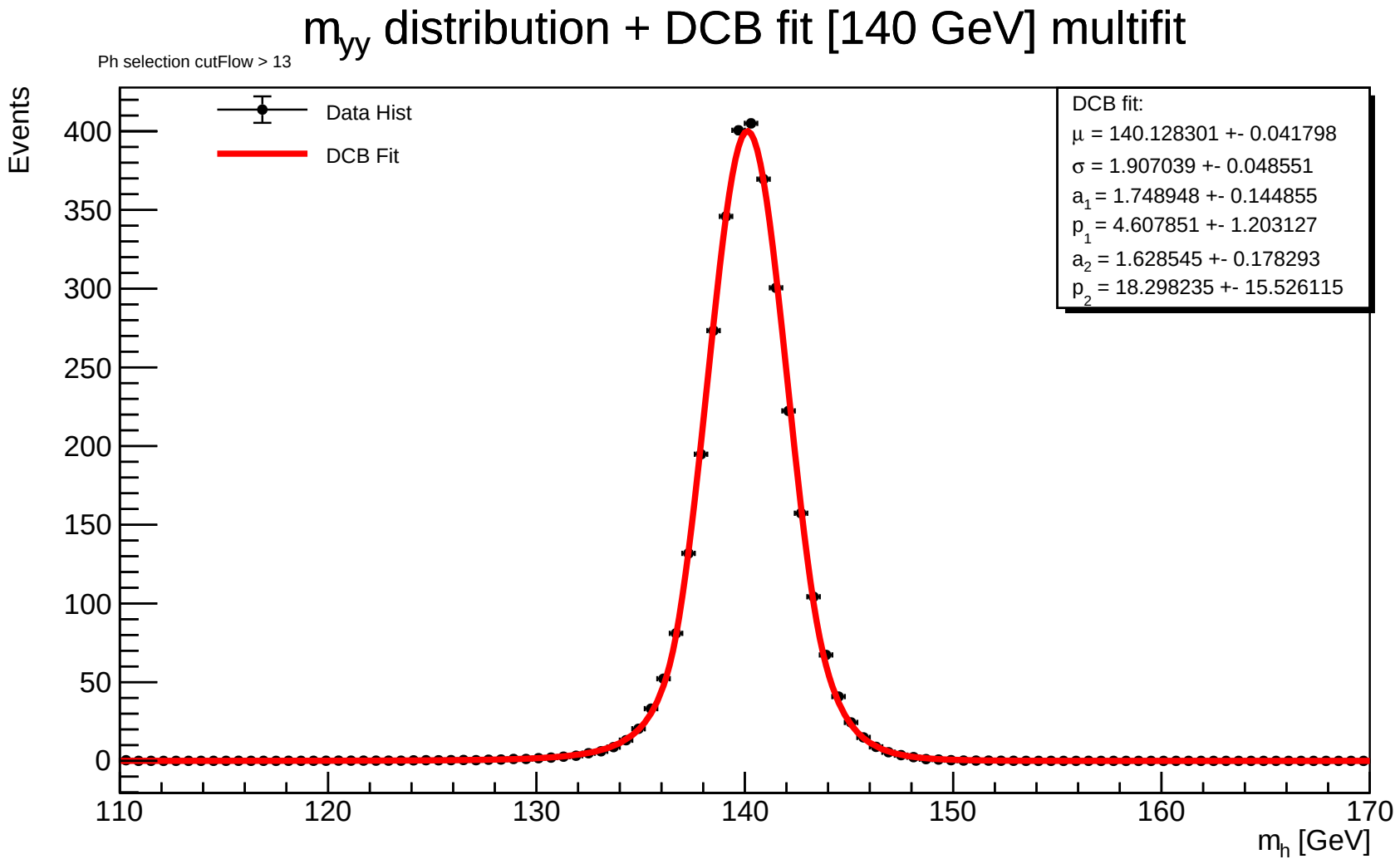
DSCB multifit [125 GeV]



DSCB multifit [130 GeV]



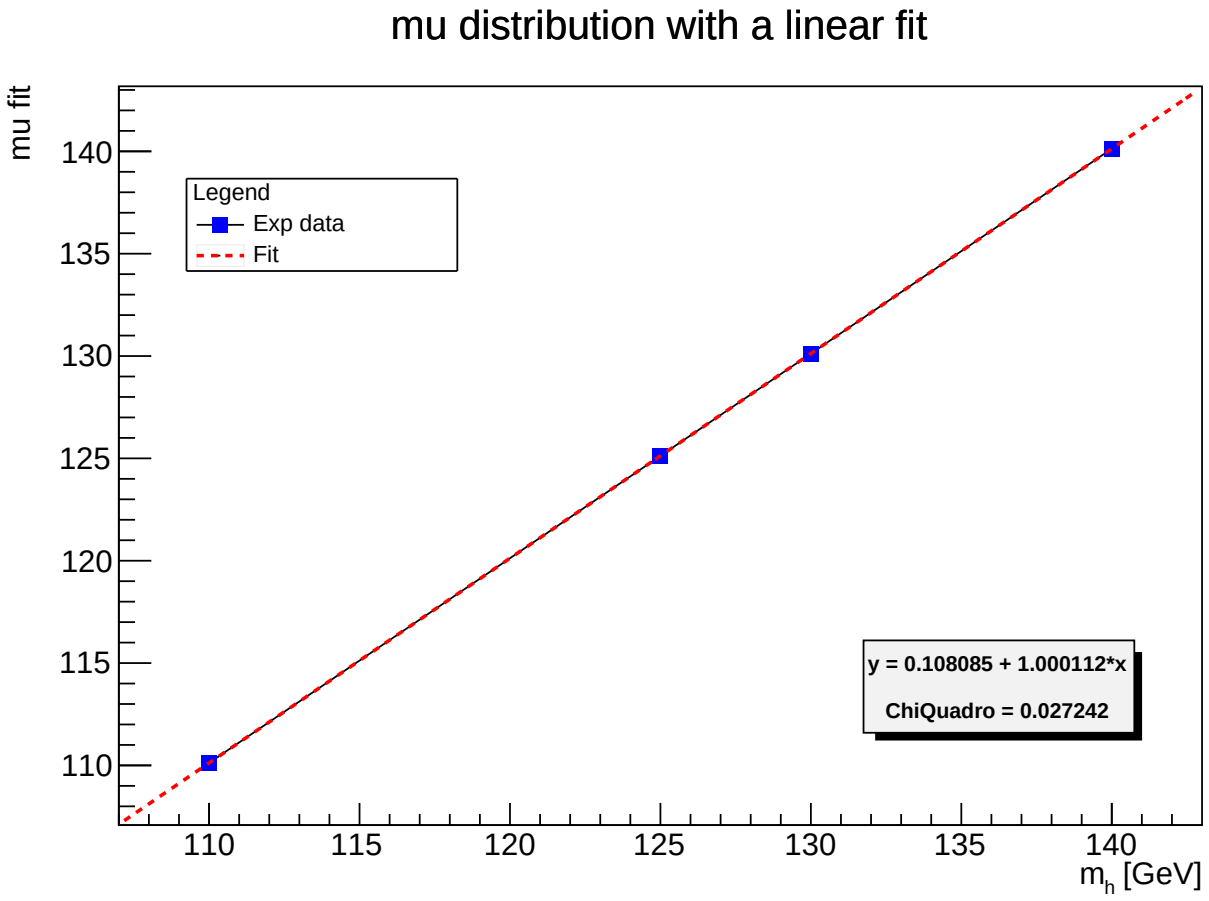
DSCB multifit [140 GeV]



$\mu(m_h)$

Fitting the μ values at different m_h with a linear fit:

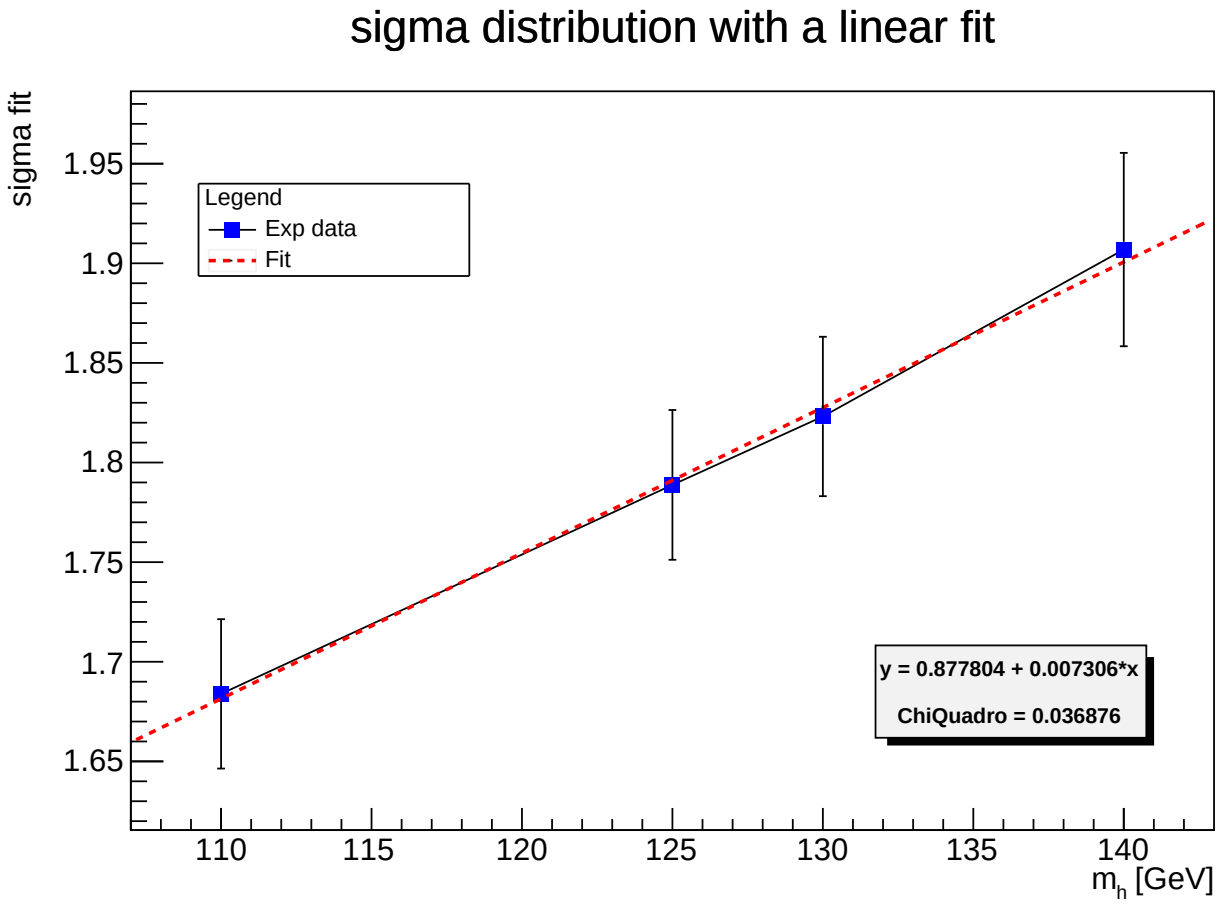
$$\mu(m_h) = A + B \cdot m_h$$



$\sigma(m_h)$

Fitting the σ values at different m_h with a linear fit:

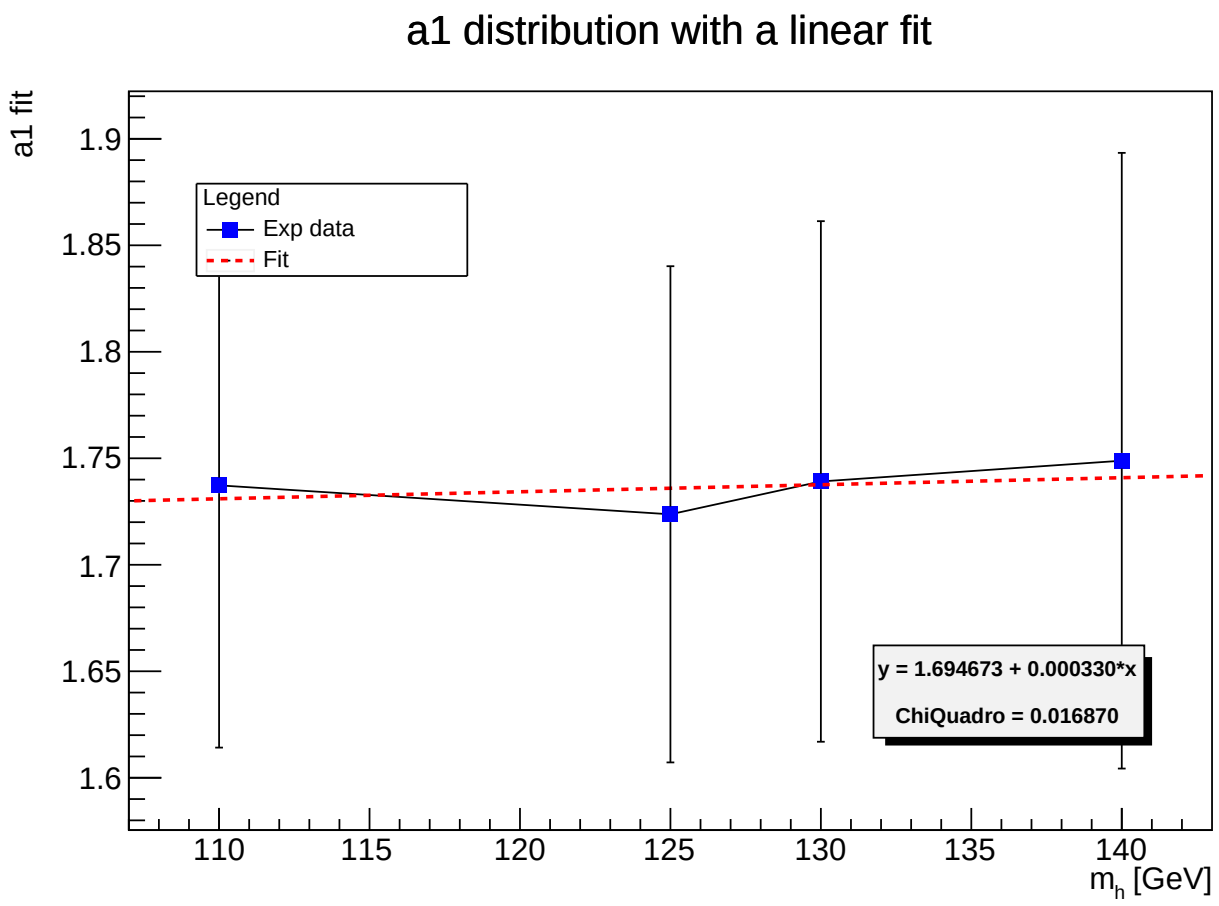
$$\sigma(m_h) = A + B \cdot m_h$$



$a_1(m_h)$

Fitting the a_1 values at different m_h with a linear fit:

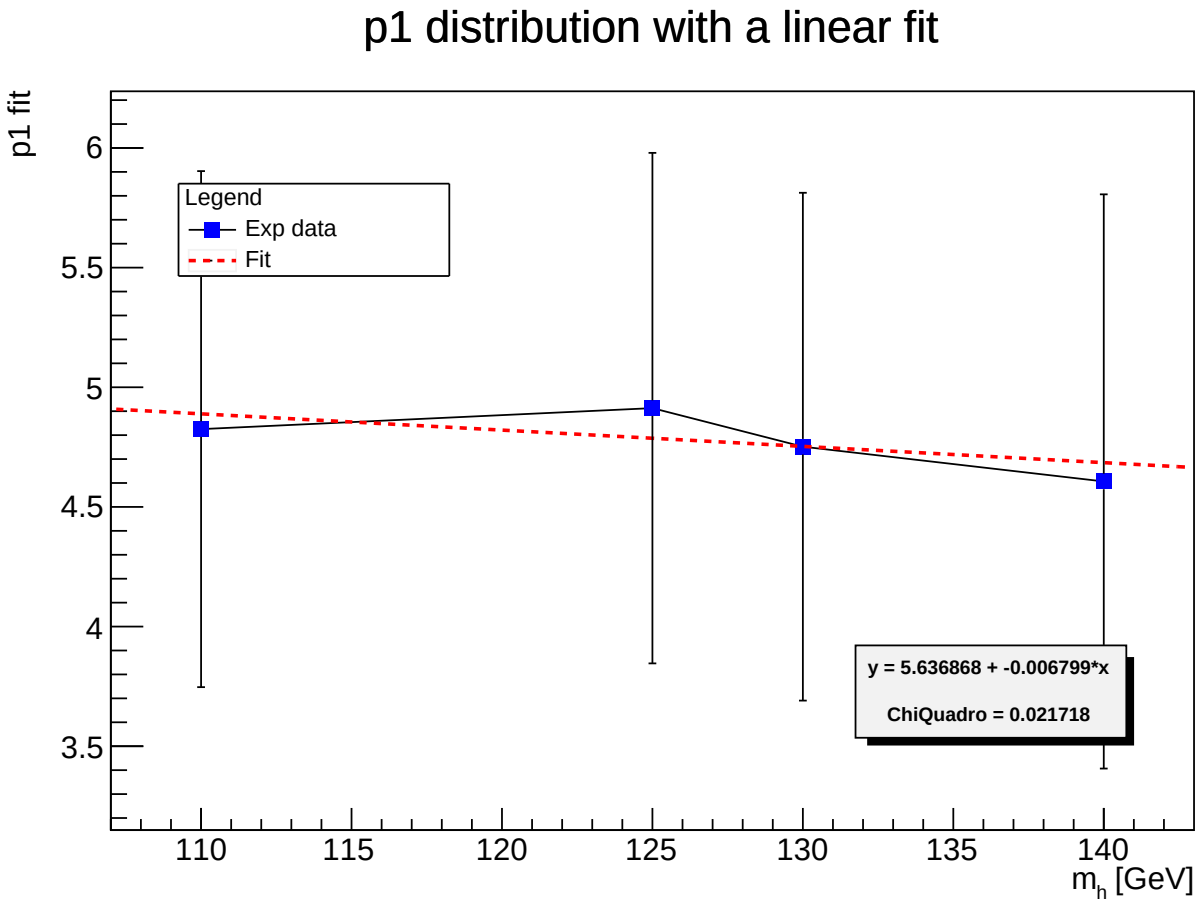
$$a_1(m_h) = A + B \cdot m_h$$



$p_1(m_h)$

Fitting the p_1 values at different m_h with a linear fit:

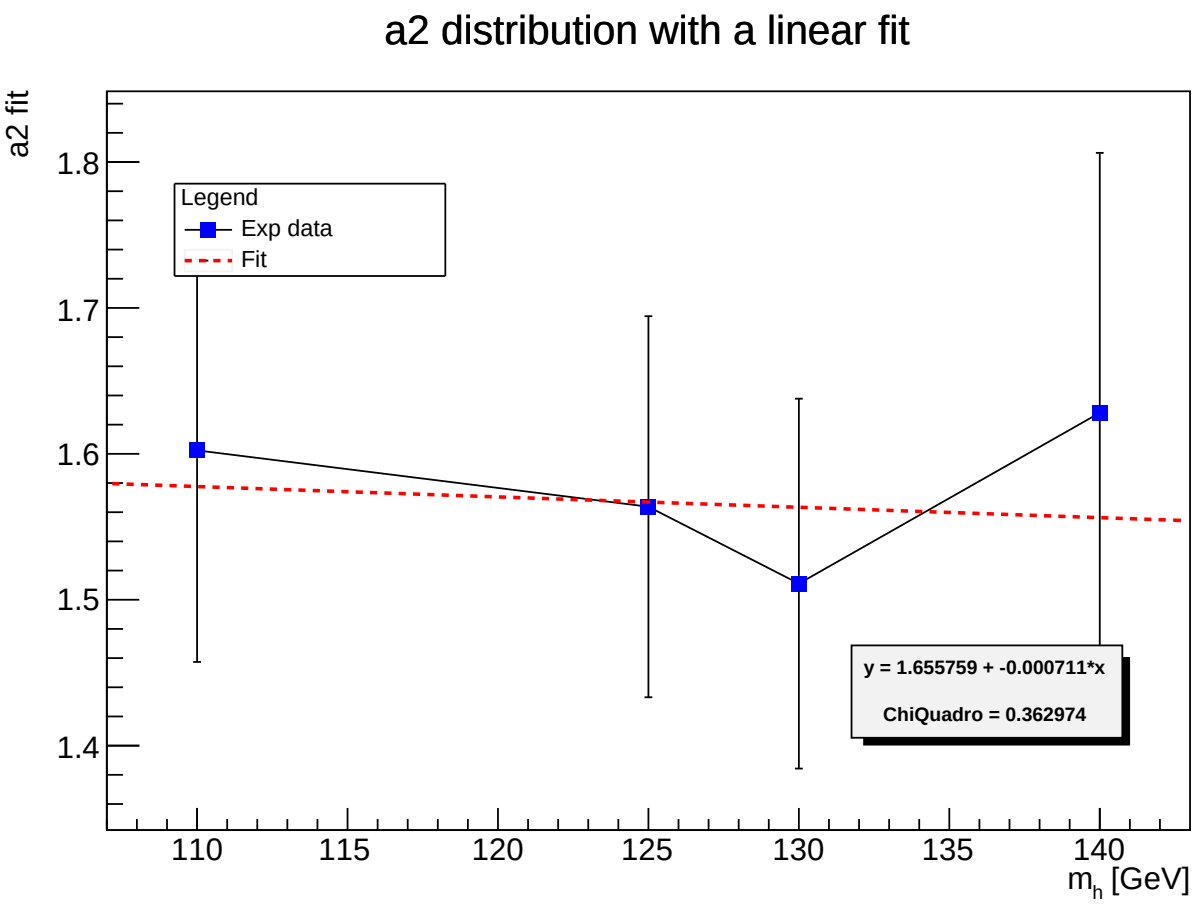
$$p_1(m_h) = A + B \cdot m_h$$



$a_2(m_h)$

Fitting the a_2 values at different m_h with a linear fit:

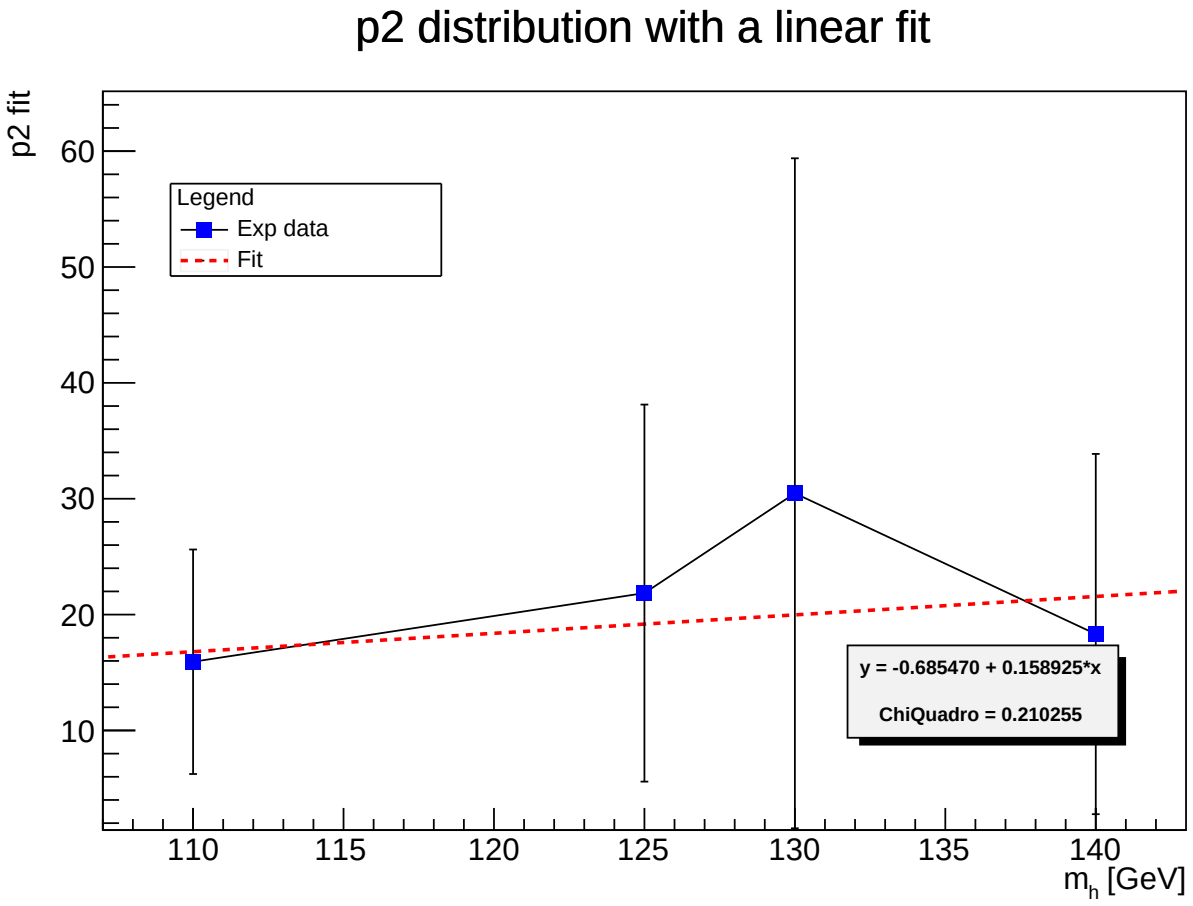
$$a_2(m_h) = A + B \cdot m_h$$



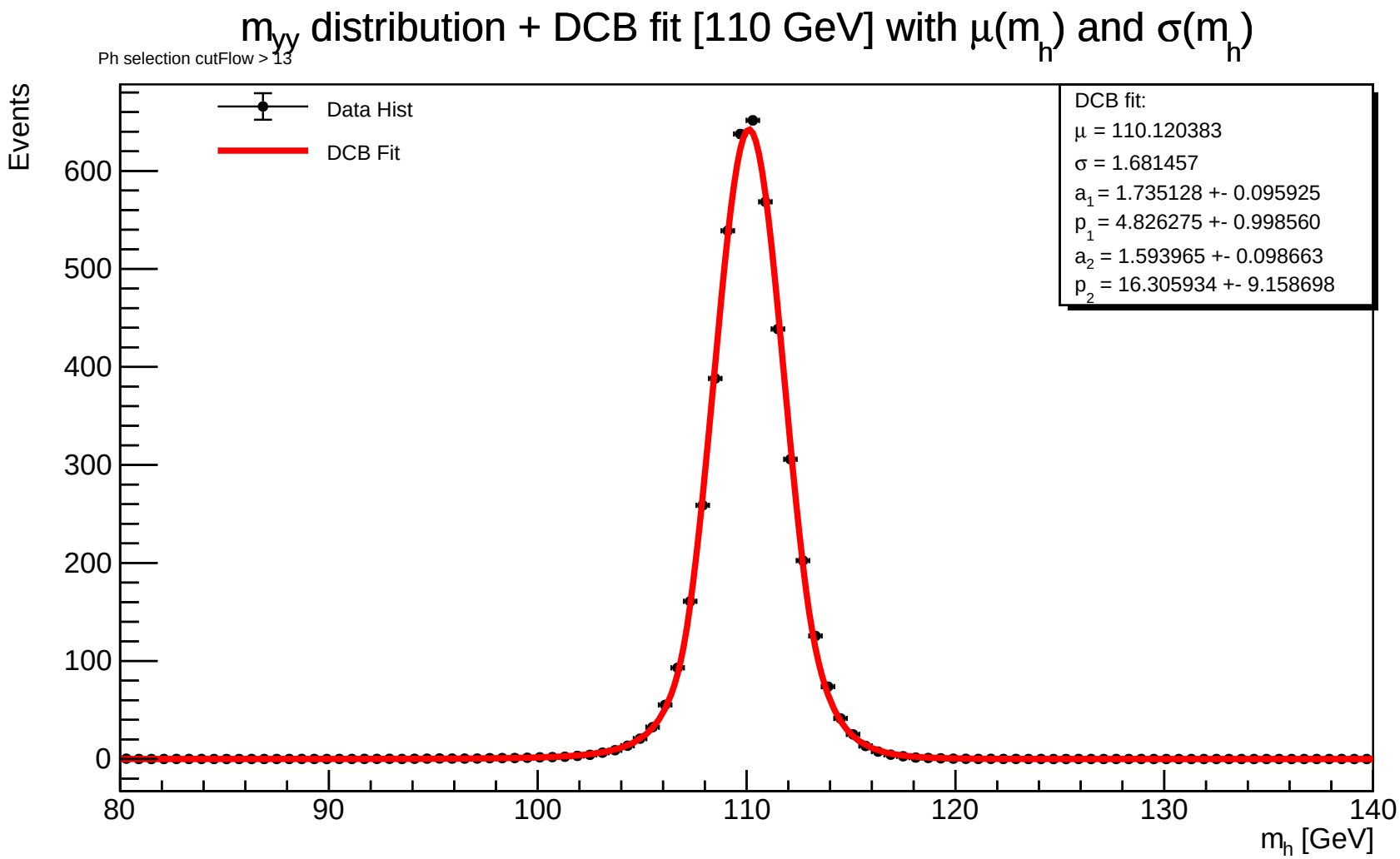
$p_2(m_h)$

Fitting the p_2 values at different m_h with a linear fit:

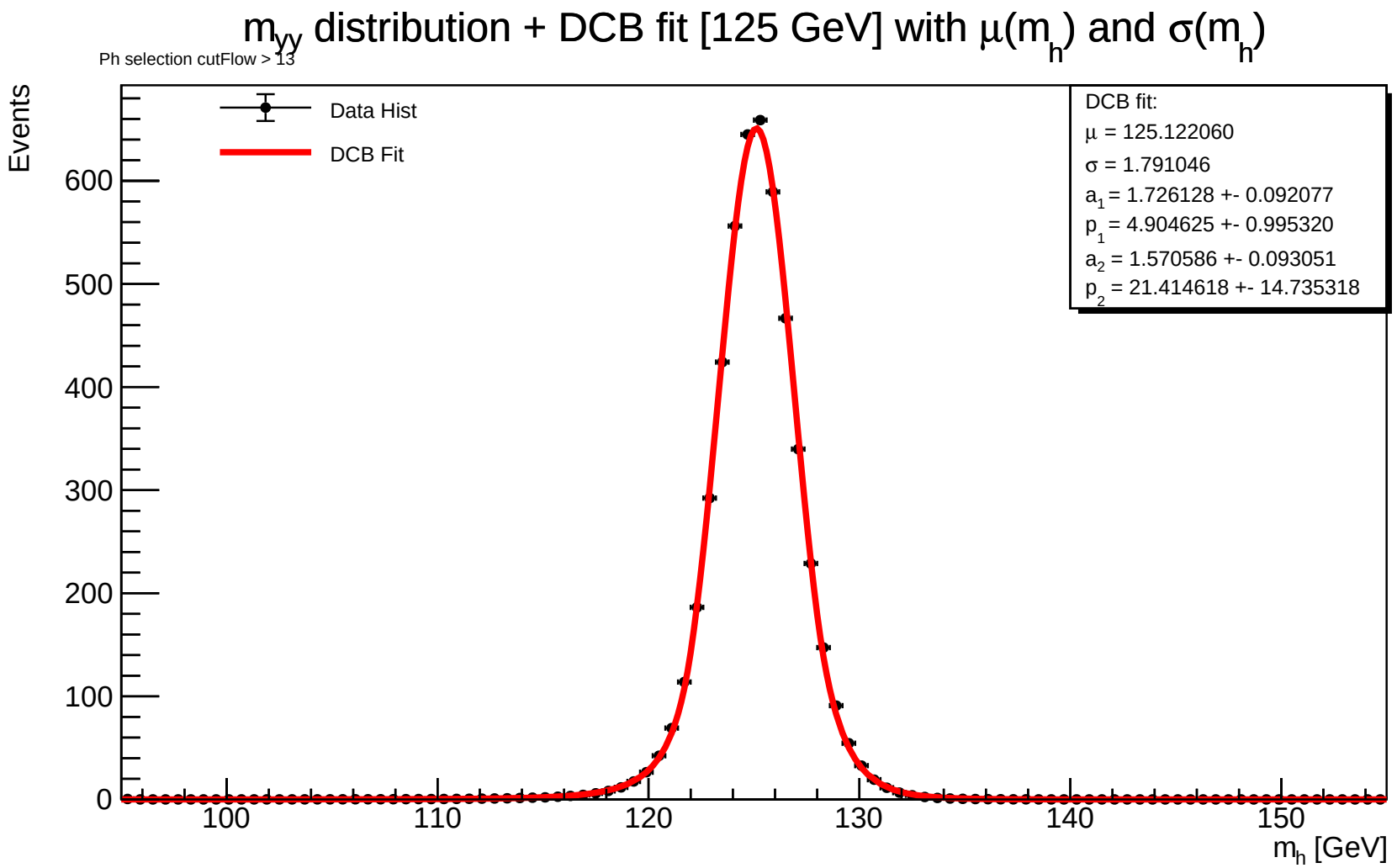
$$p_2(m_h) = A + B \cdot m_h$$



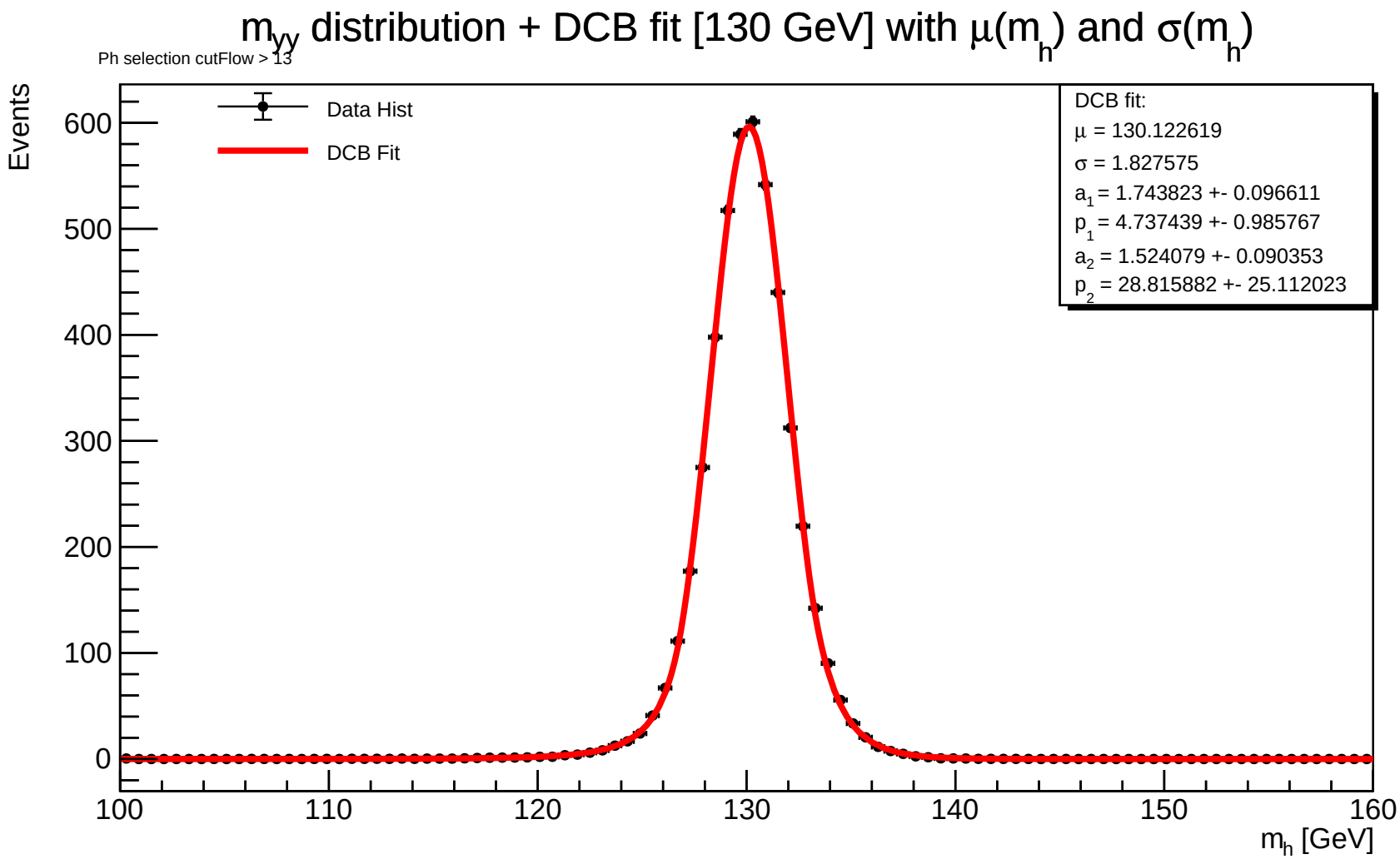
DSCB(m_h) with $\mu(m_h)$ and $\sigma(m_h)$ [110 GeV]



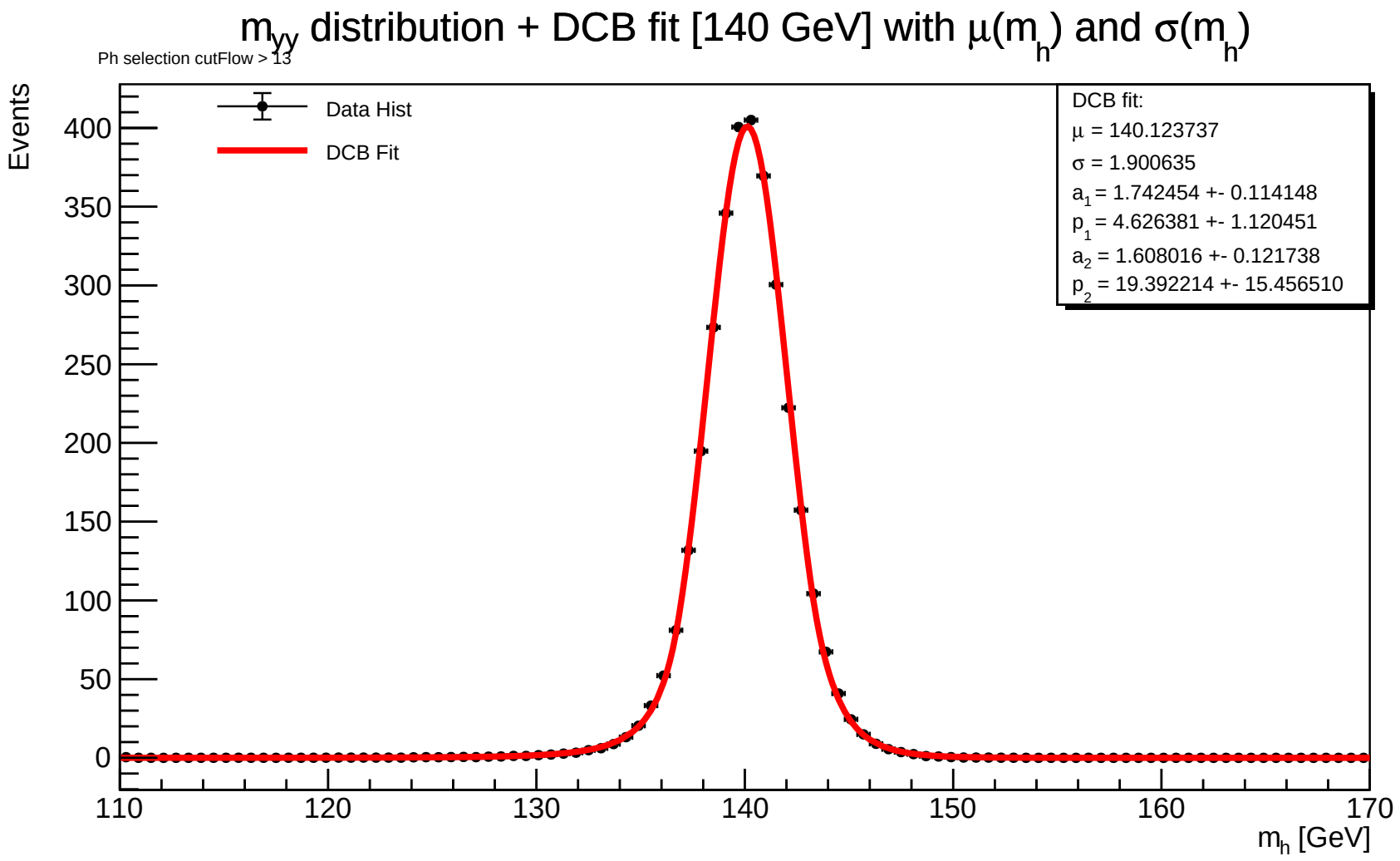
DSCB(m_h) with $\mu(m_h)$ and $\sigma(m_h)$ [125 GeV]



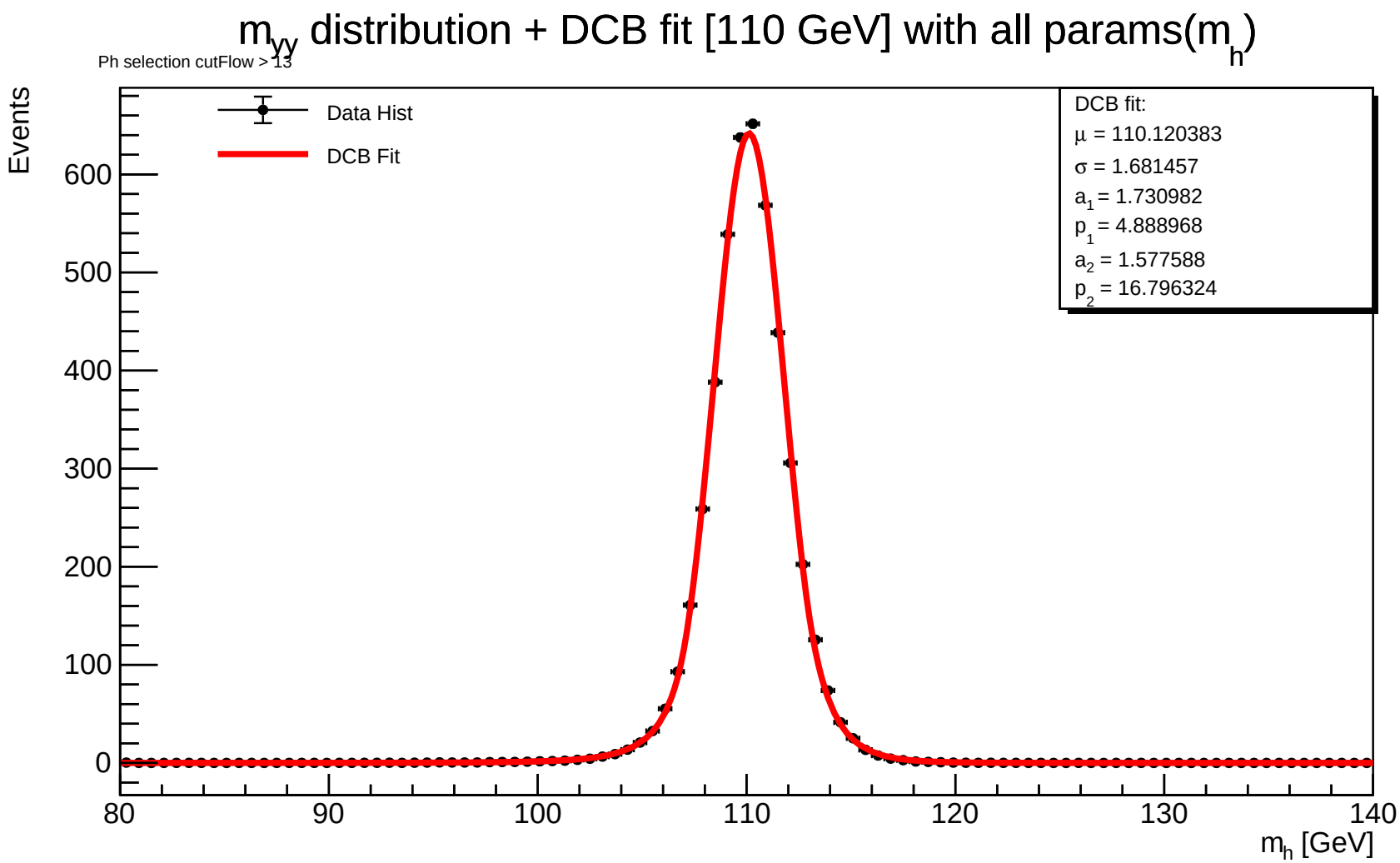
DSCB(m_h) with $\mu(m_h)$ and $\sigma(m_h)$ [130 GeV]



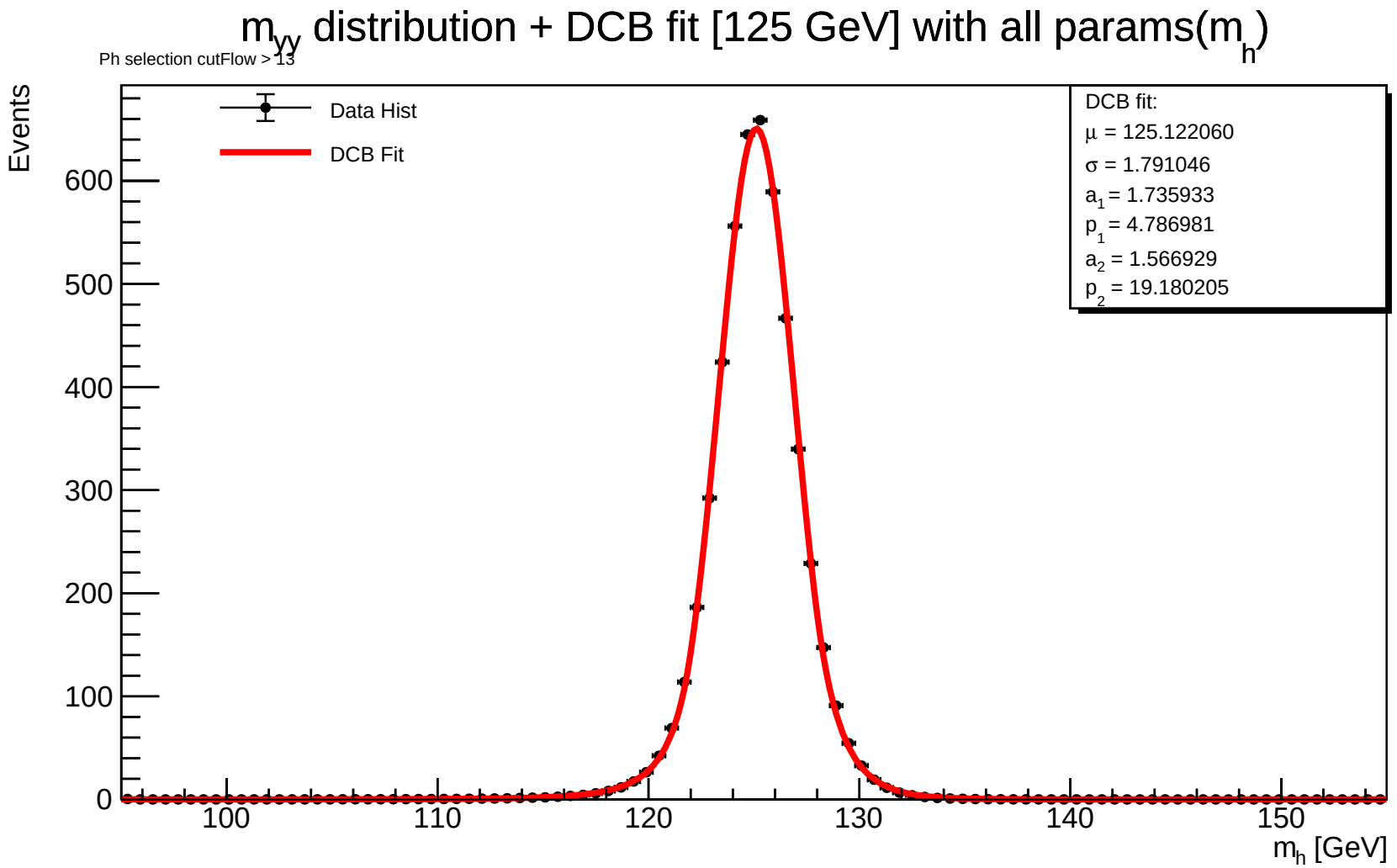
DSCB(m_h) with $\mu(m_h)$ and $\sigma(m_h)$ [140 GeV]



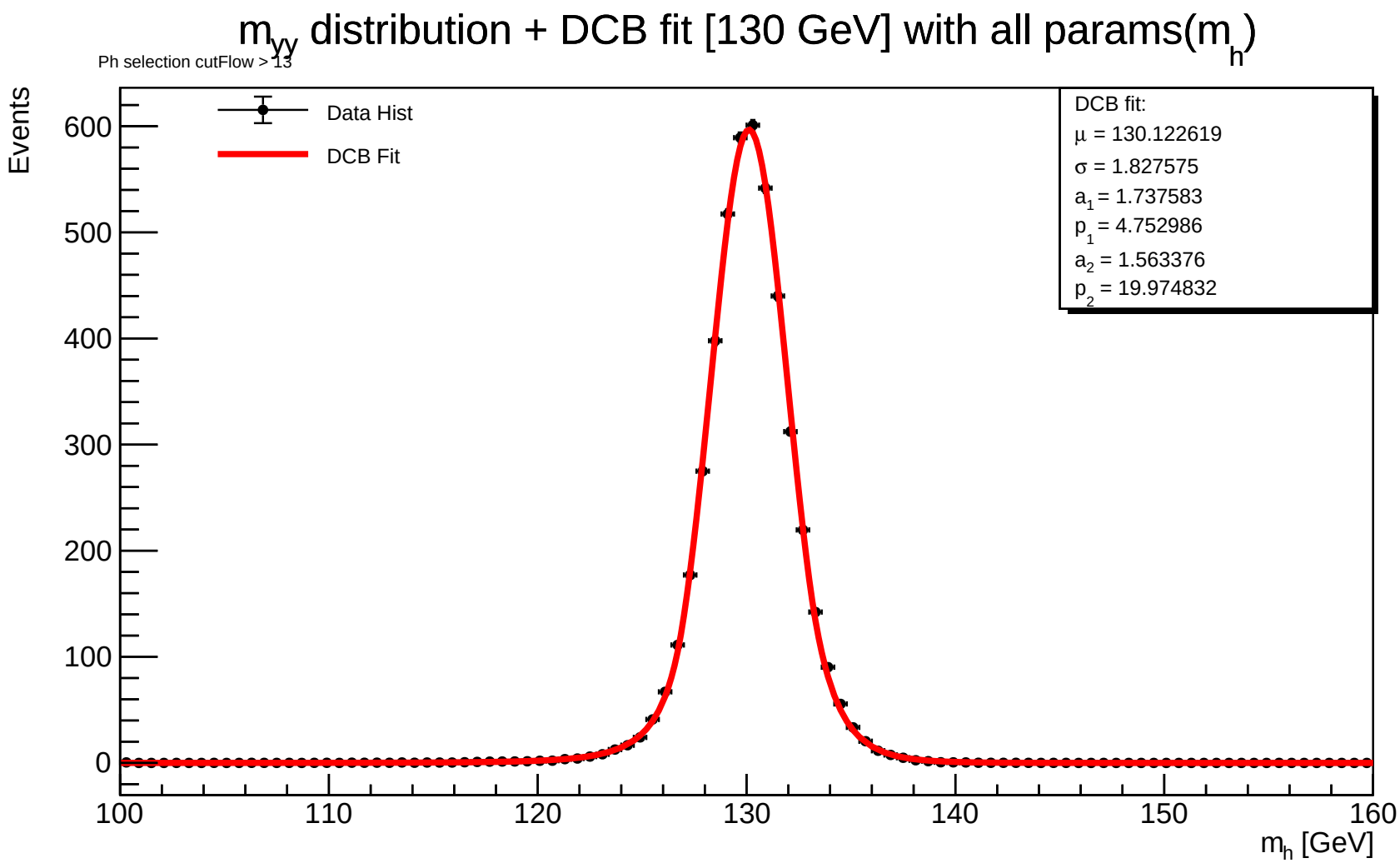
DSCB(m_h) with all params(m_h) [110 GeV]



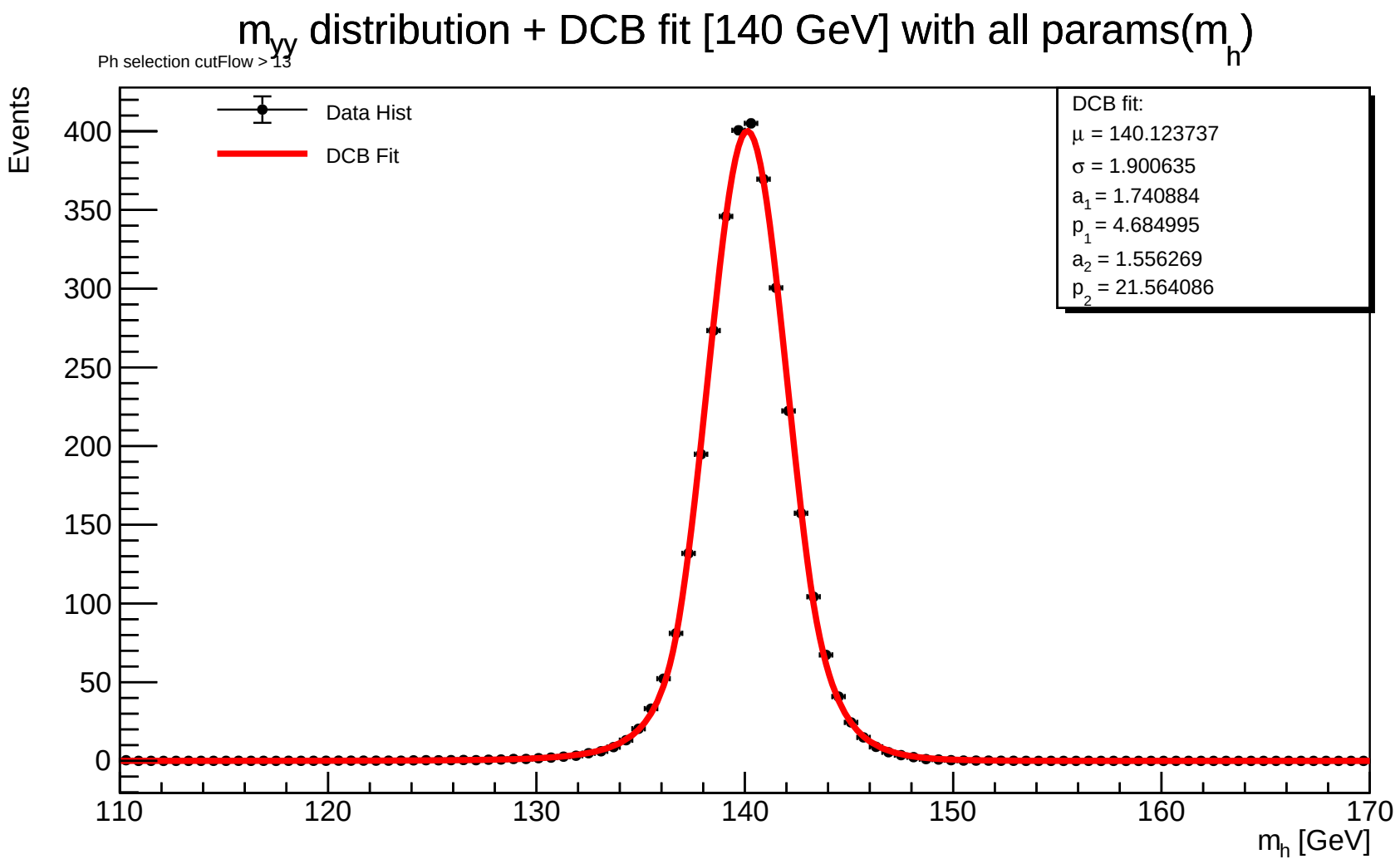
DSCB(m_h) with all params(m_h) [125 GeV]



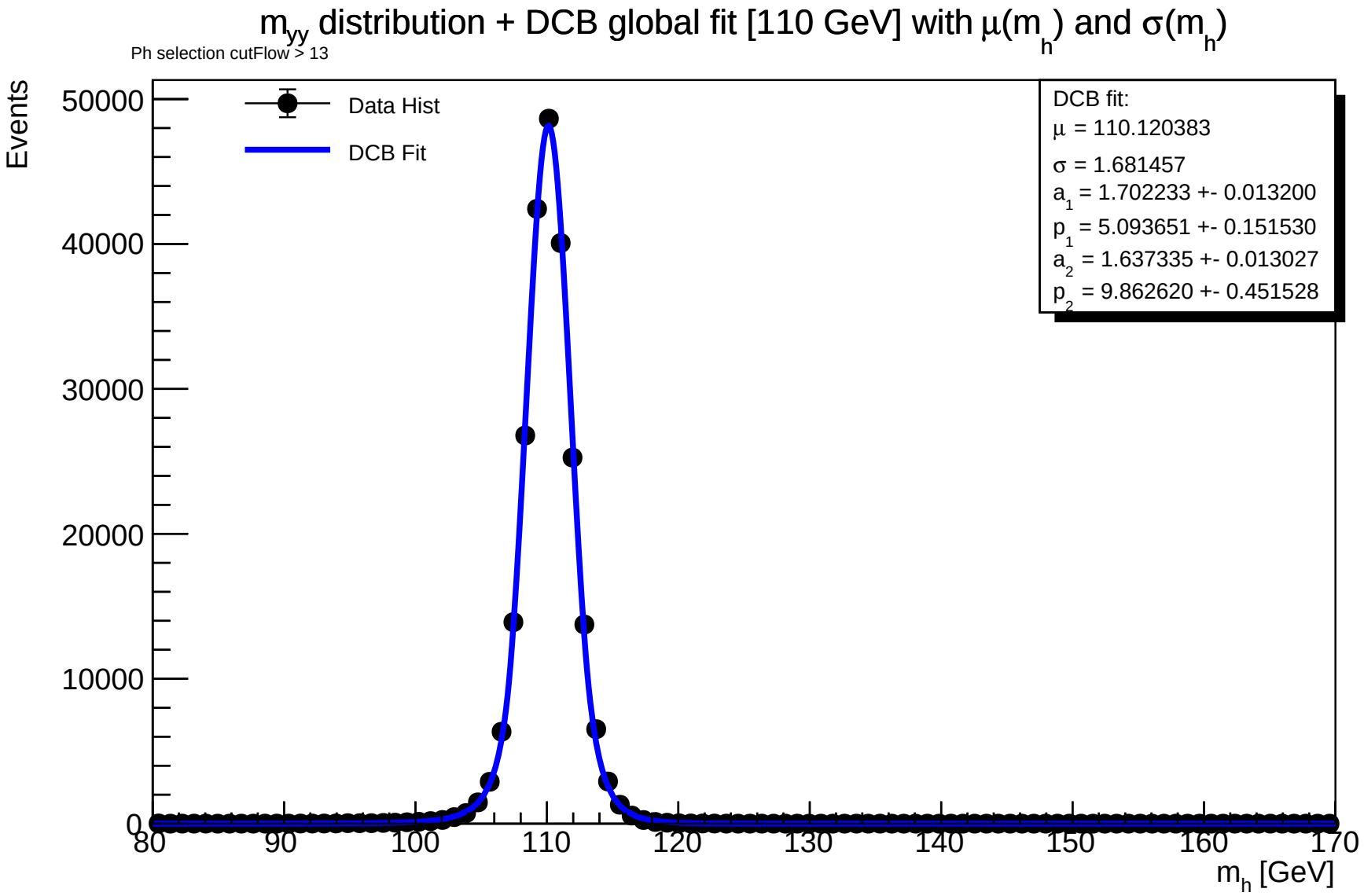
DSCB(m_h) with all params(m_h) [130 GeV]



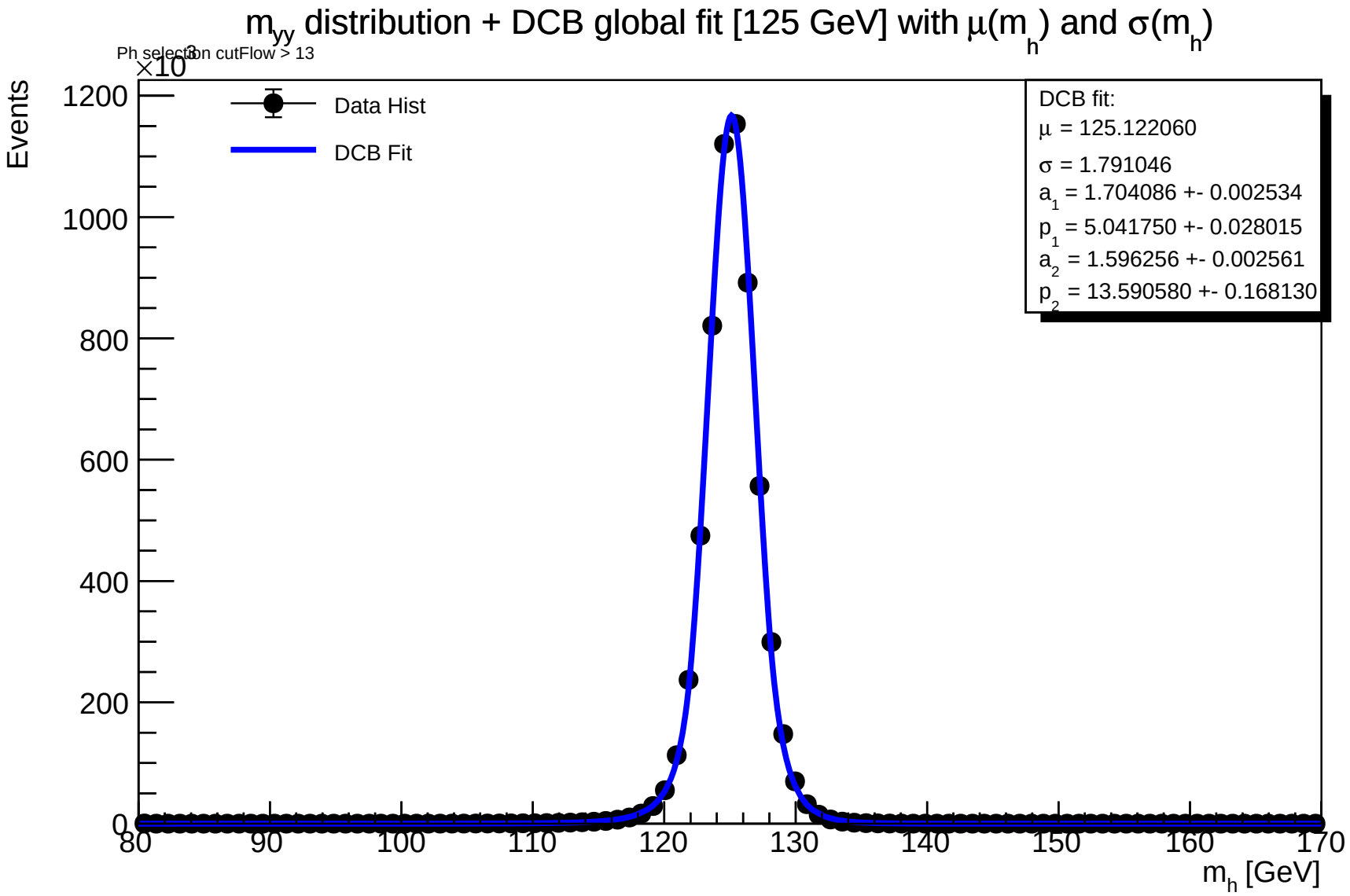
DSCB(m_h) with all params(m_h) [140 GeV]



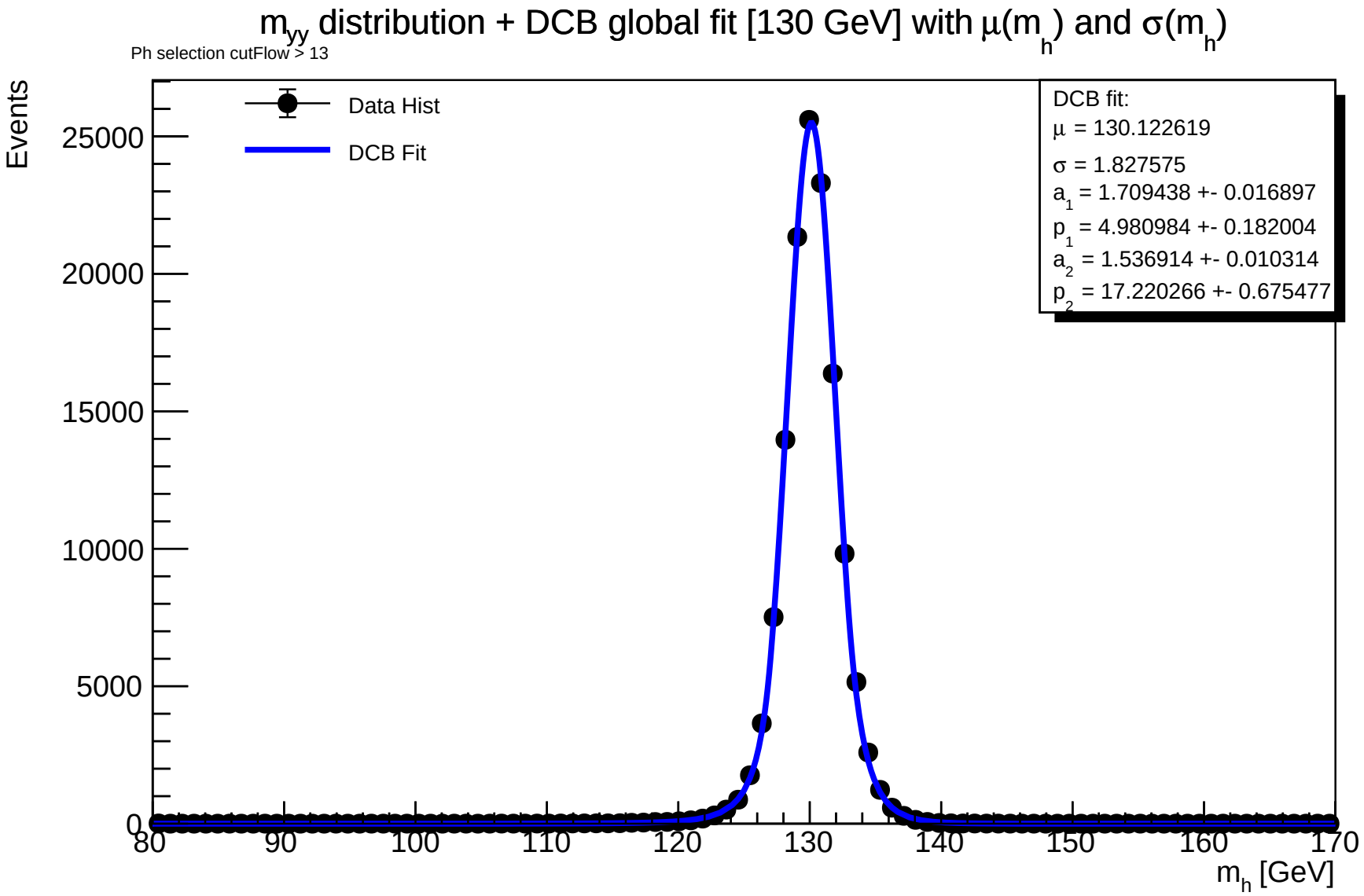
Global fit [110 GeV]



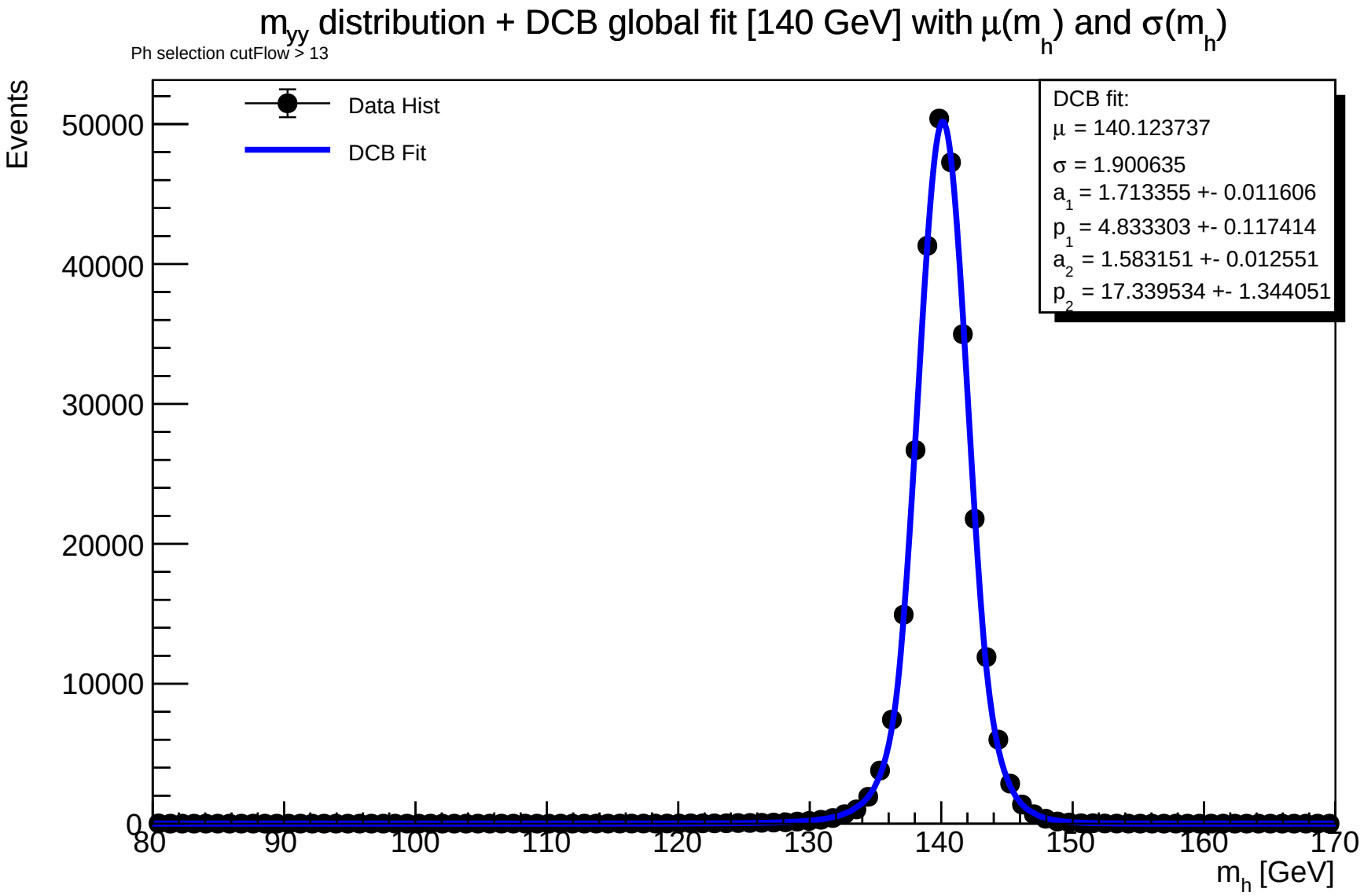
Global fit [125 GeV]



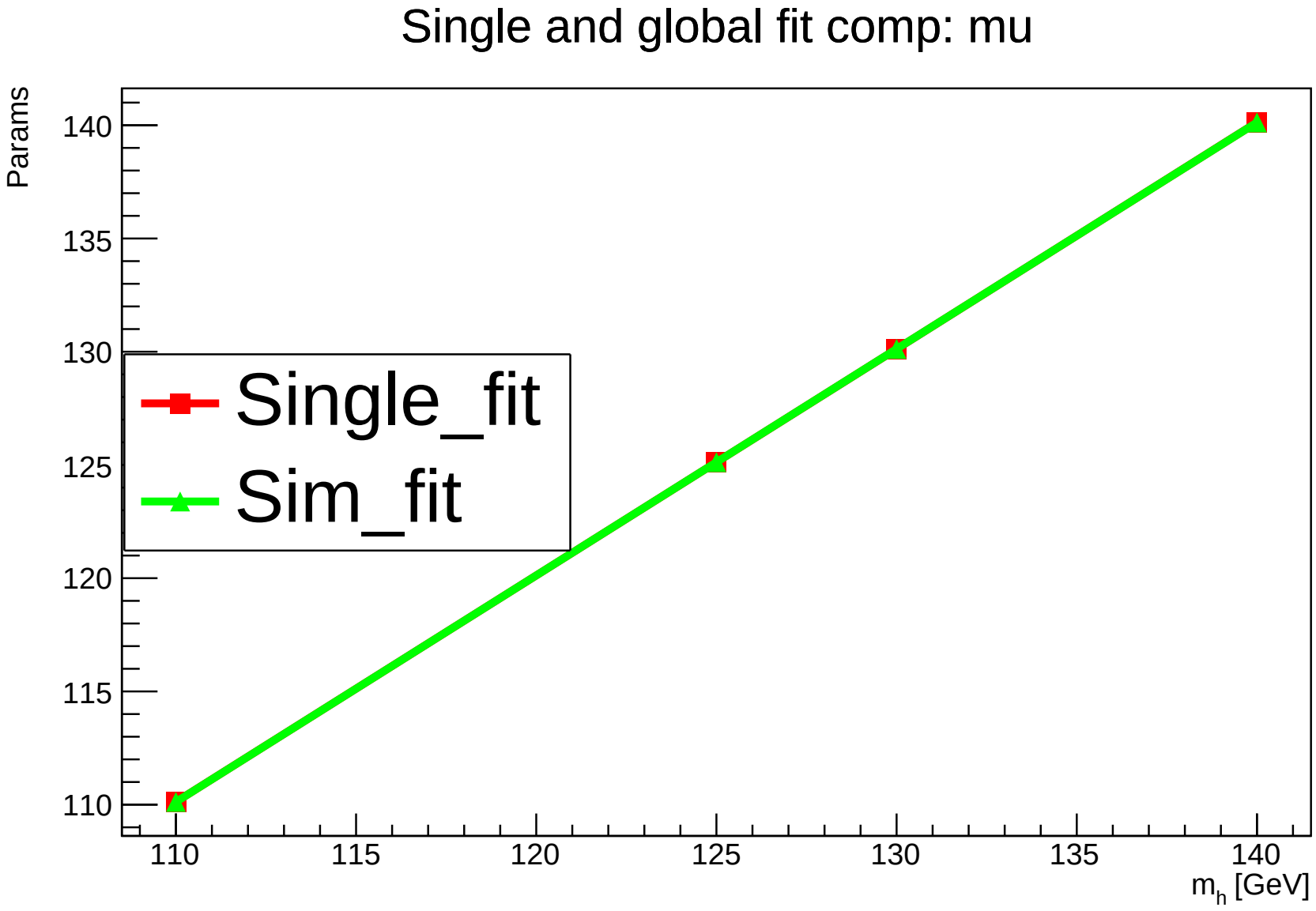
Global fit [130 GeV]



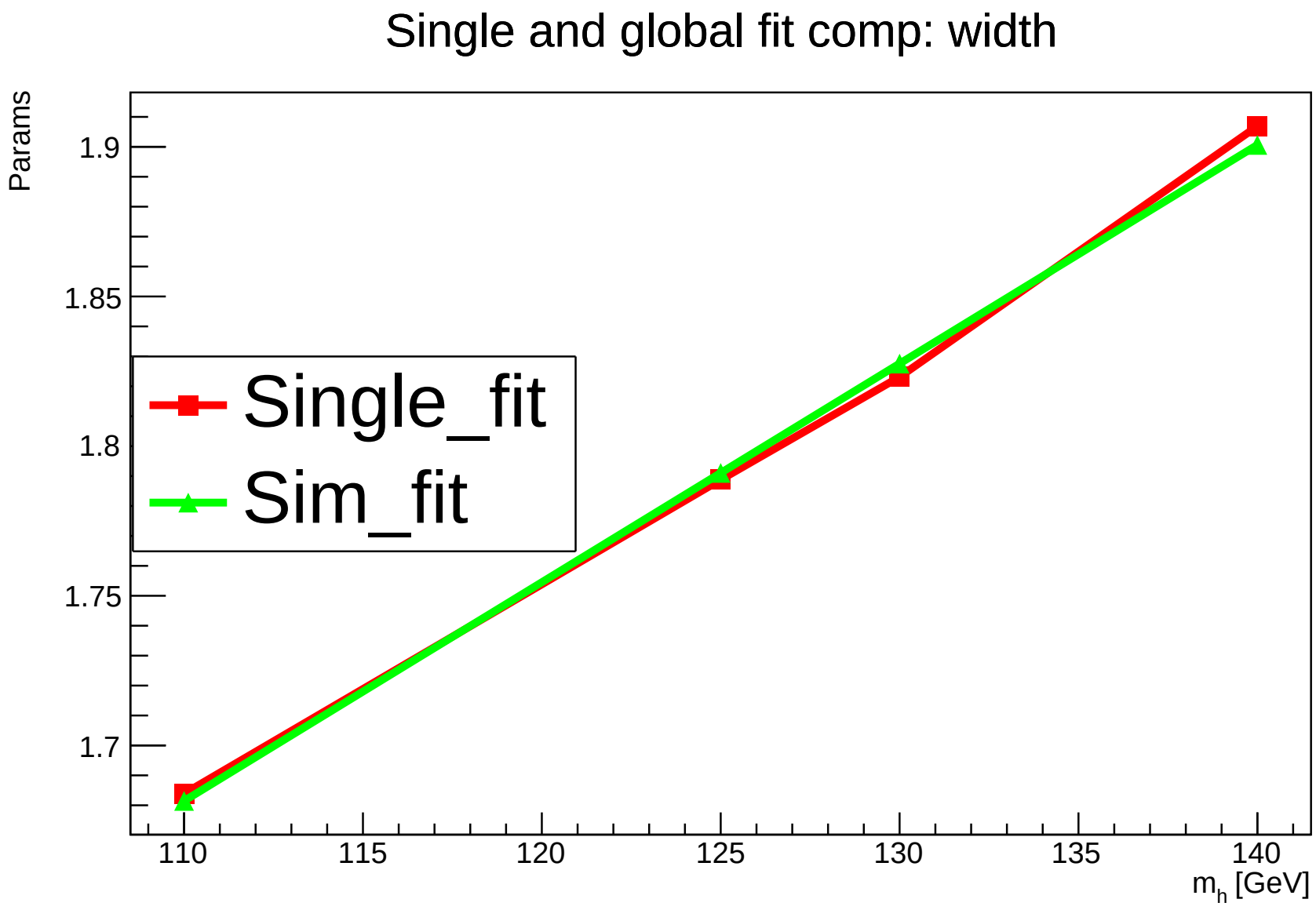
Global fit [140 GeV]



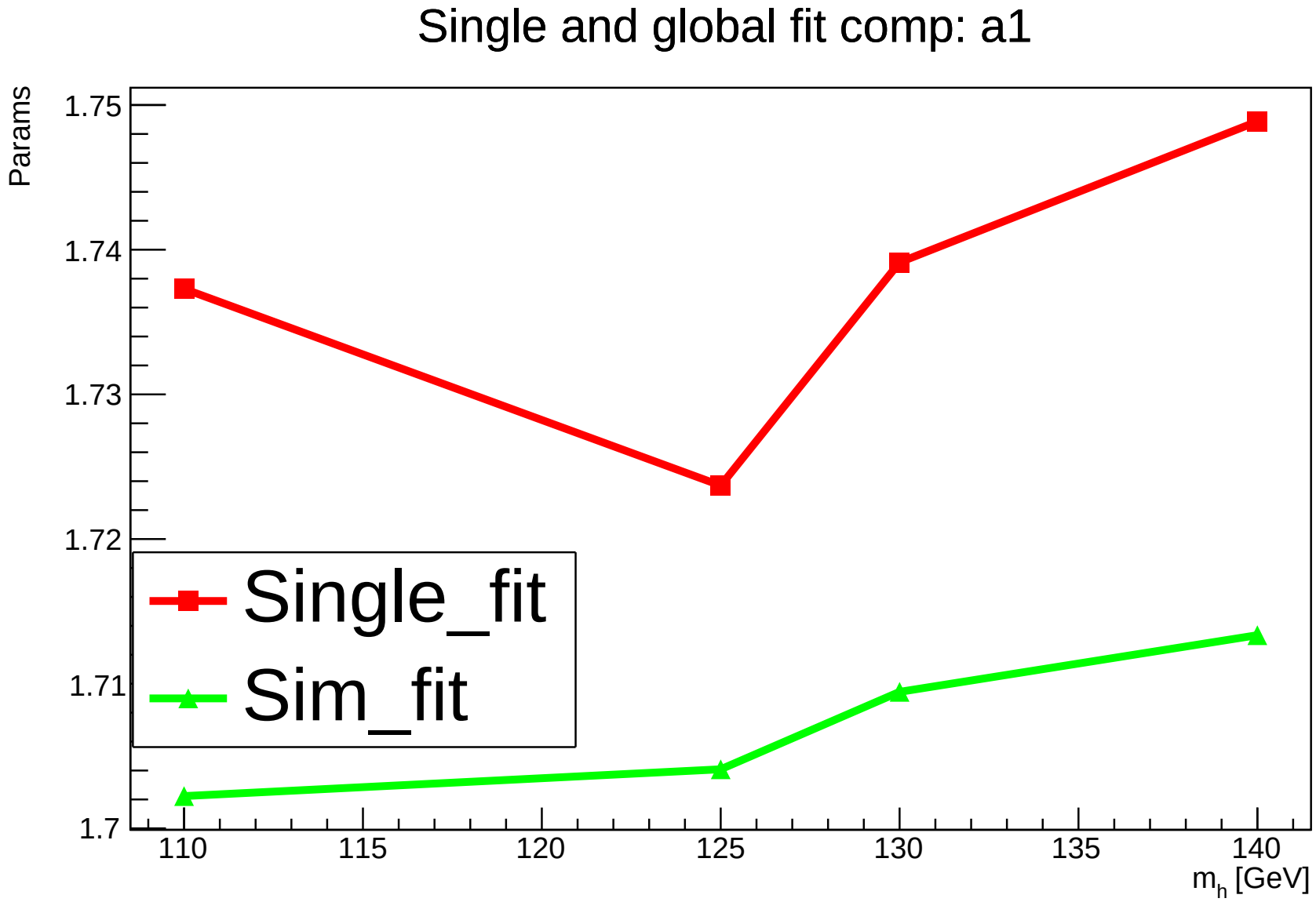
Single vs Global fit params: μ



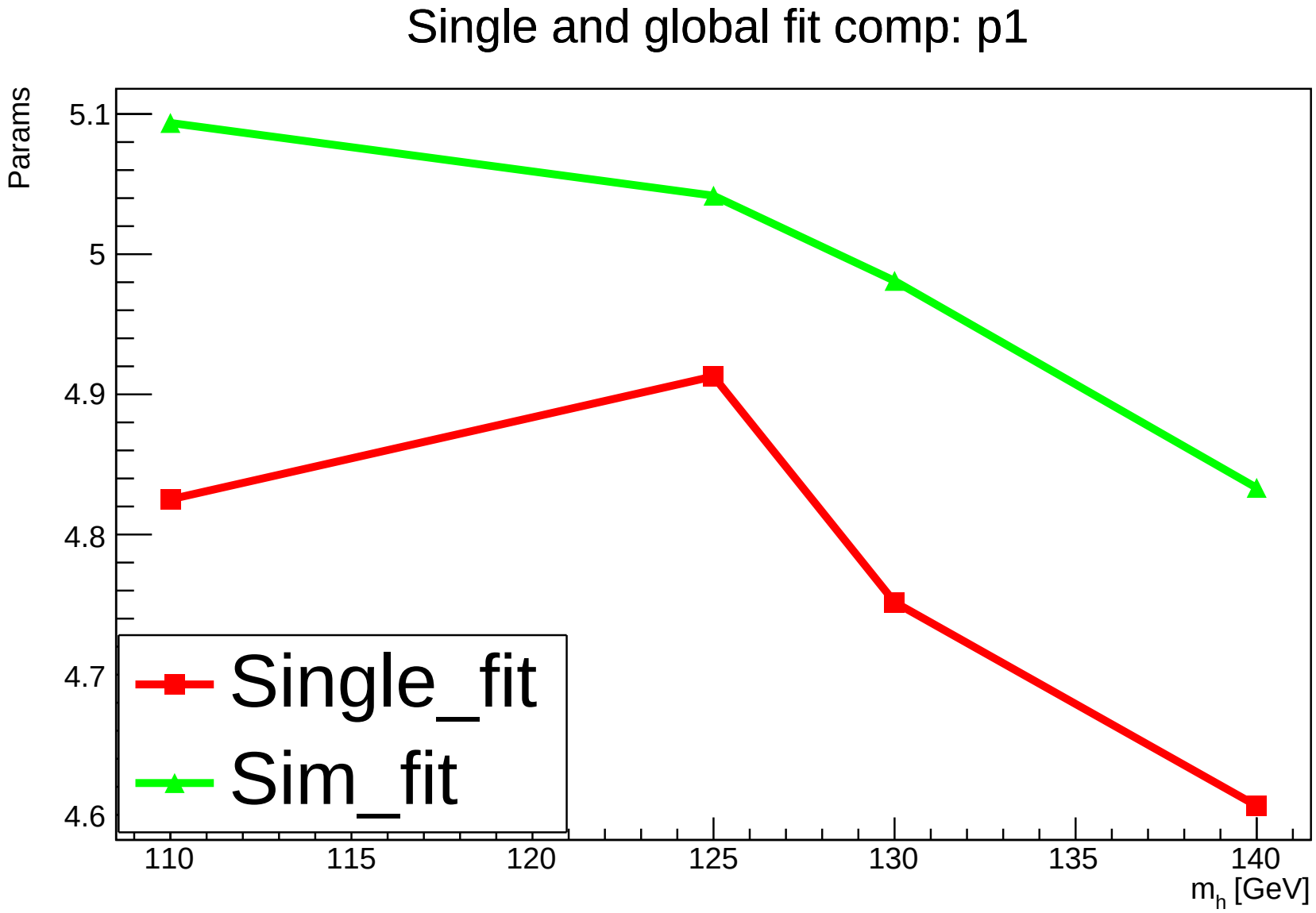
Single vs Global fit params: σ



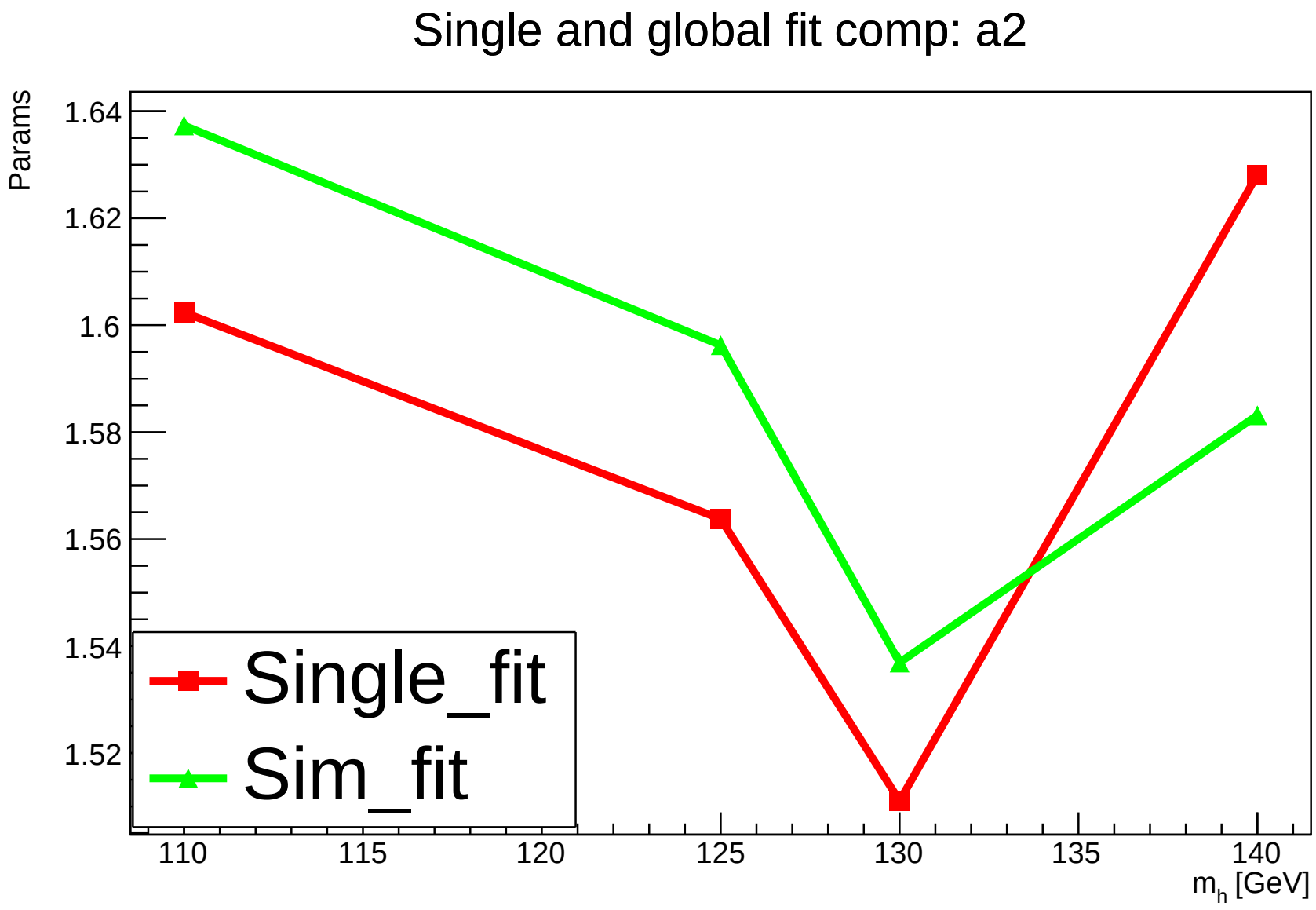
Single vs Global fit params: a_1



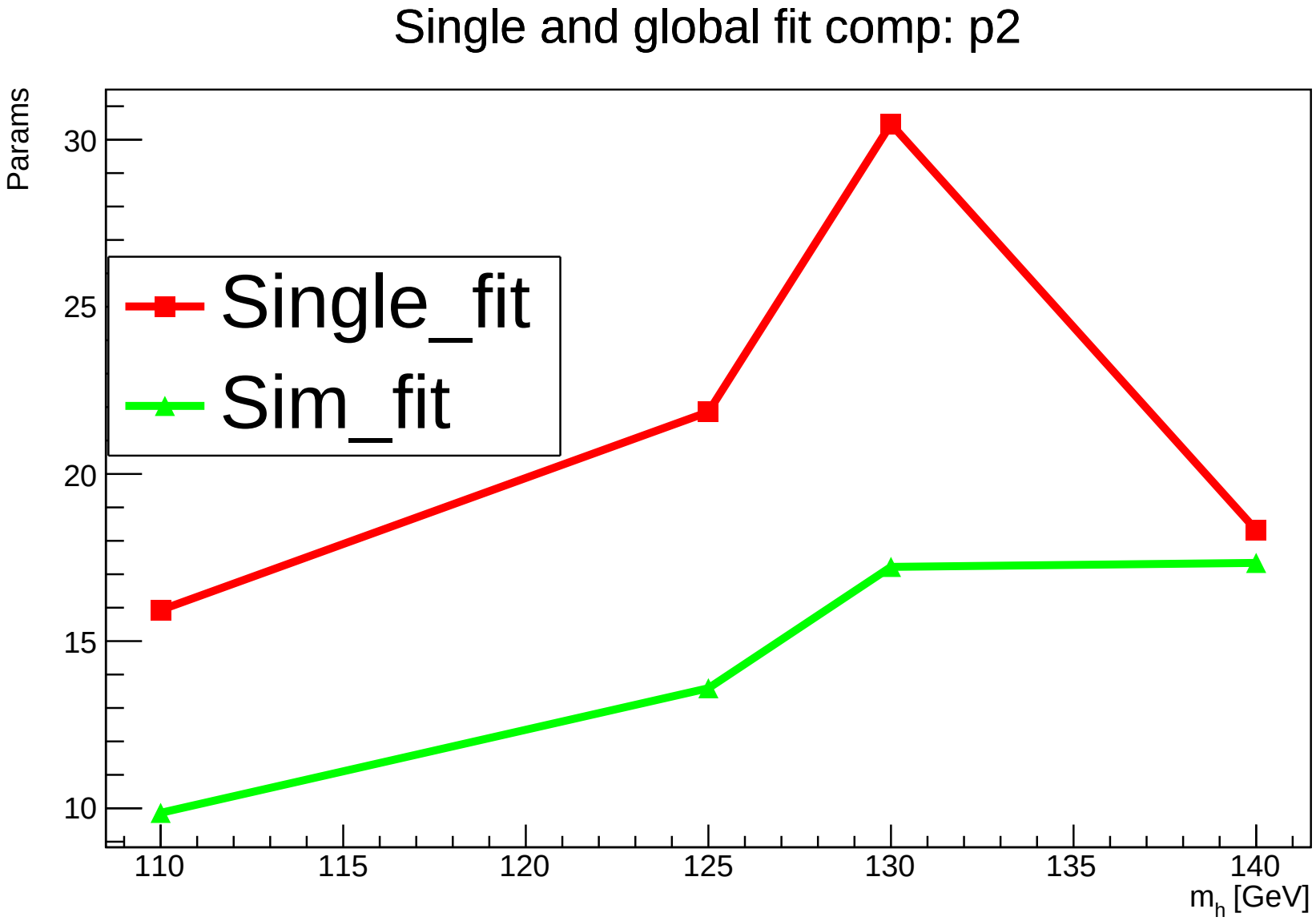
Single vs Global fit params: p_1



Single vs Global fit params: a_2

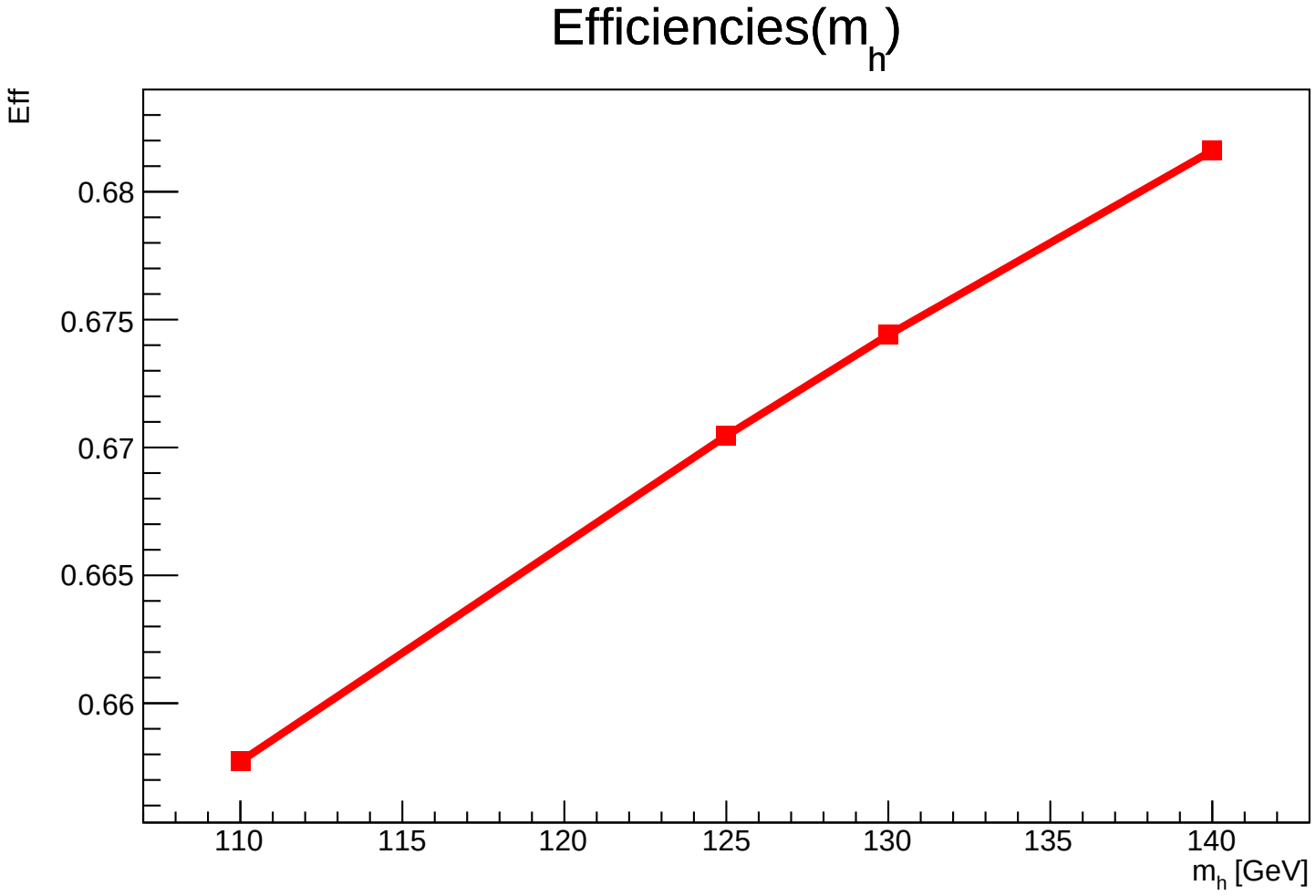


Single vs Global fit params: p_2



Efficiencies(m_h)

$$eff = \frac{\#\ of\ events\ with\ cutFlow > 13}{\#\ of\ all\ events}$$

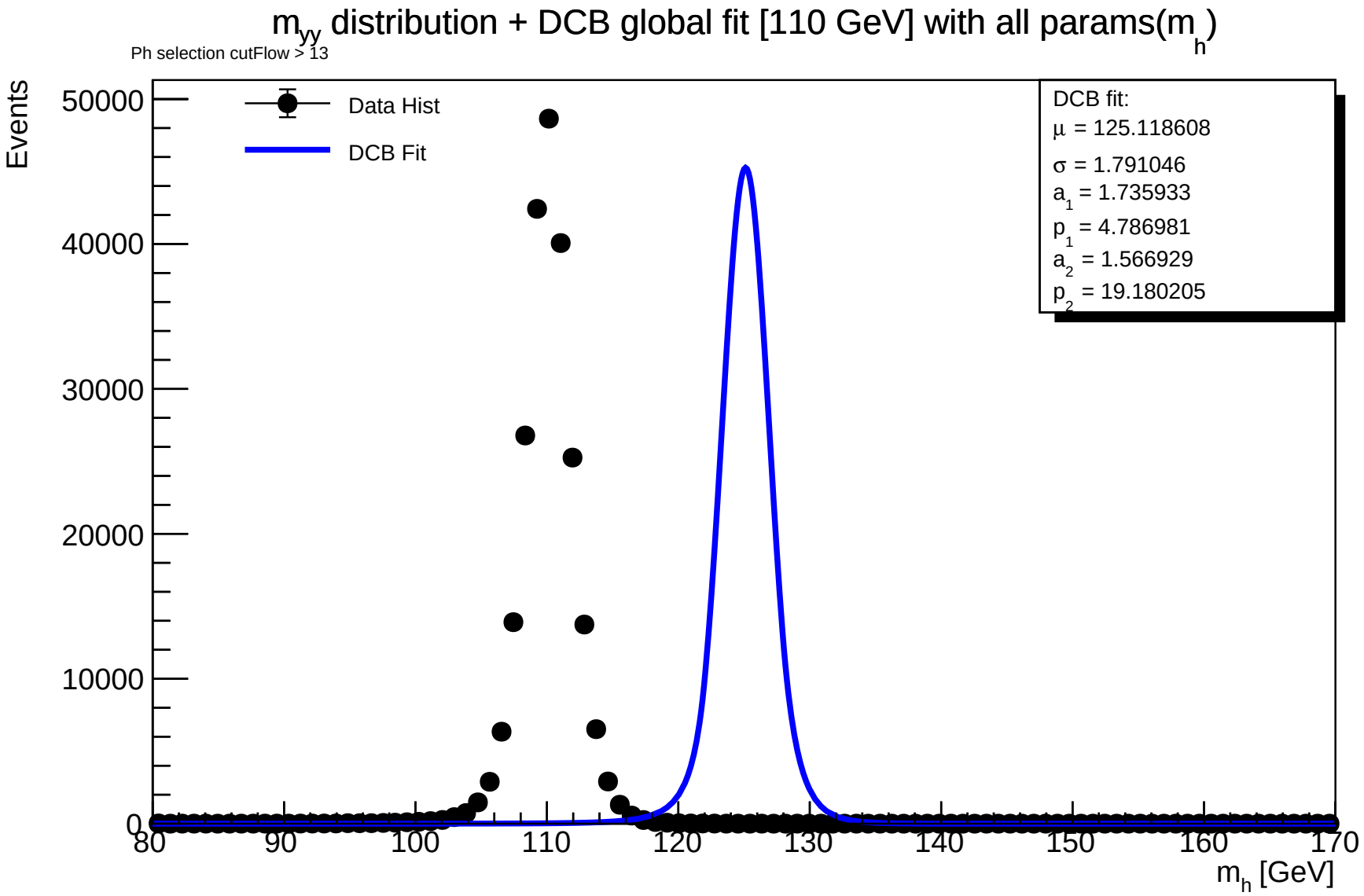


3° week

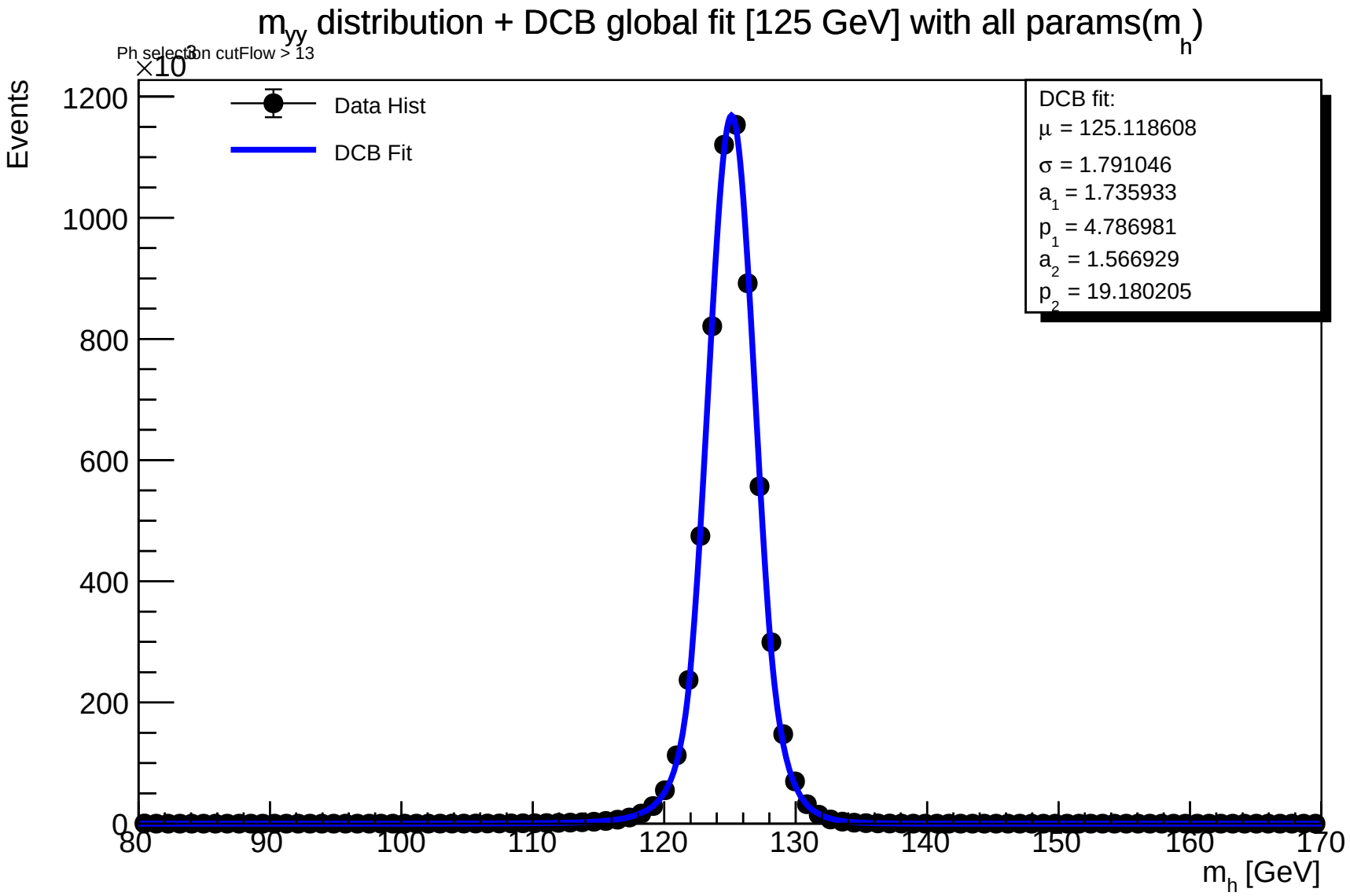
3° week assignments

- fix the X-axis:" m_{yy} $[GeV]$" to "m_h $[GeV]$";
- create a fit with all parameters as function of m_H;
- fit without a dataset:
  - the fit is build using 110, 130, 140 GeV datasets;
  - once the is created, it is applied to the 125 dataset.
- add errors bars;
- background MC:
  - plot the mass distribution;
  - apply a exp fit.
- build a p_0 scan on MC;

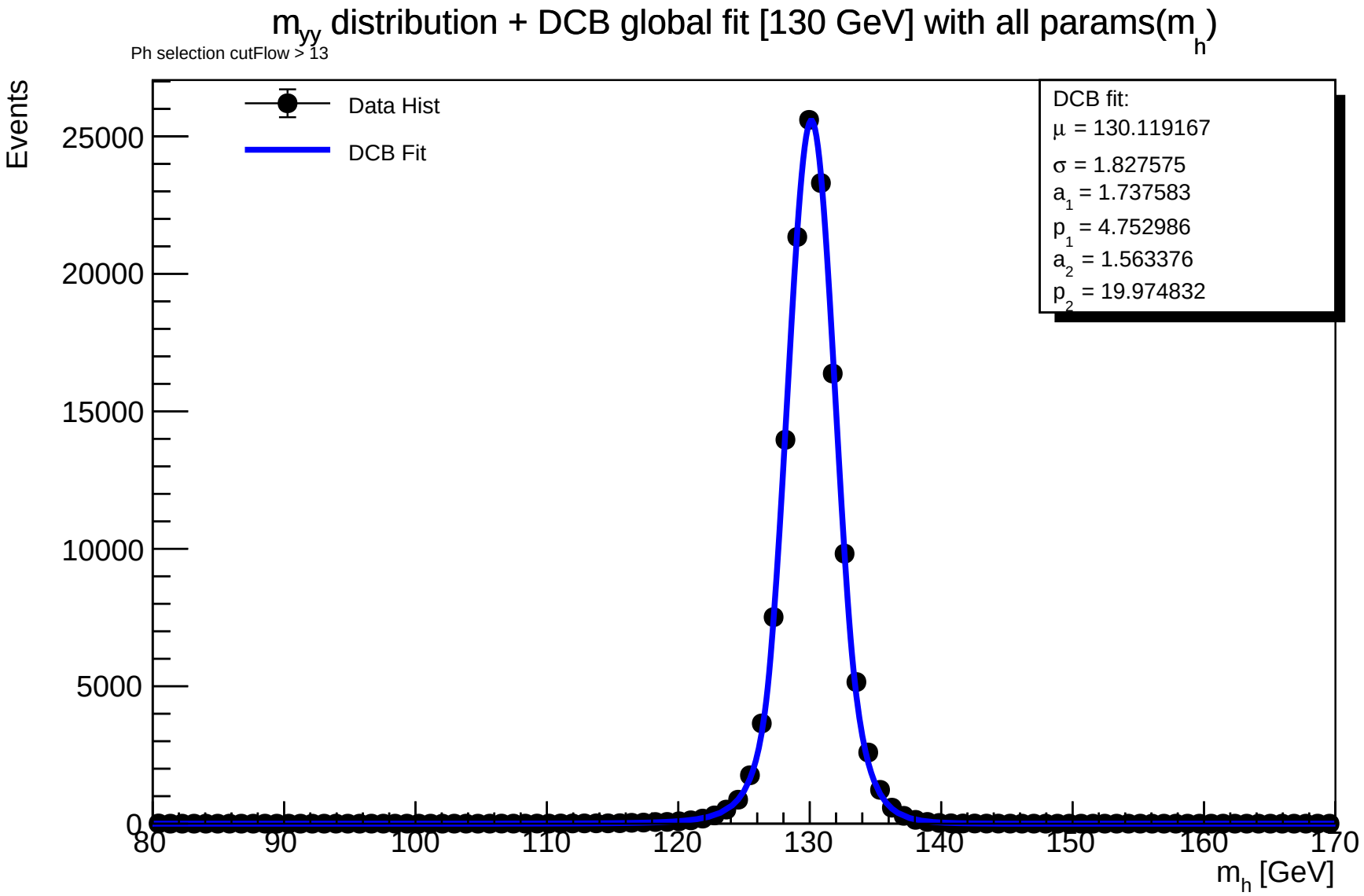
Global fit all params(m_H) [110 GeV]



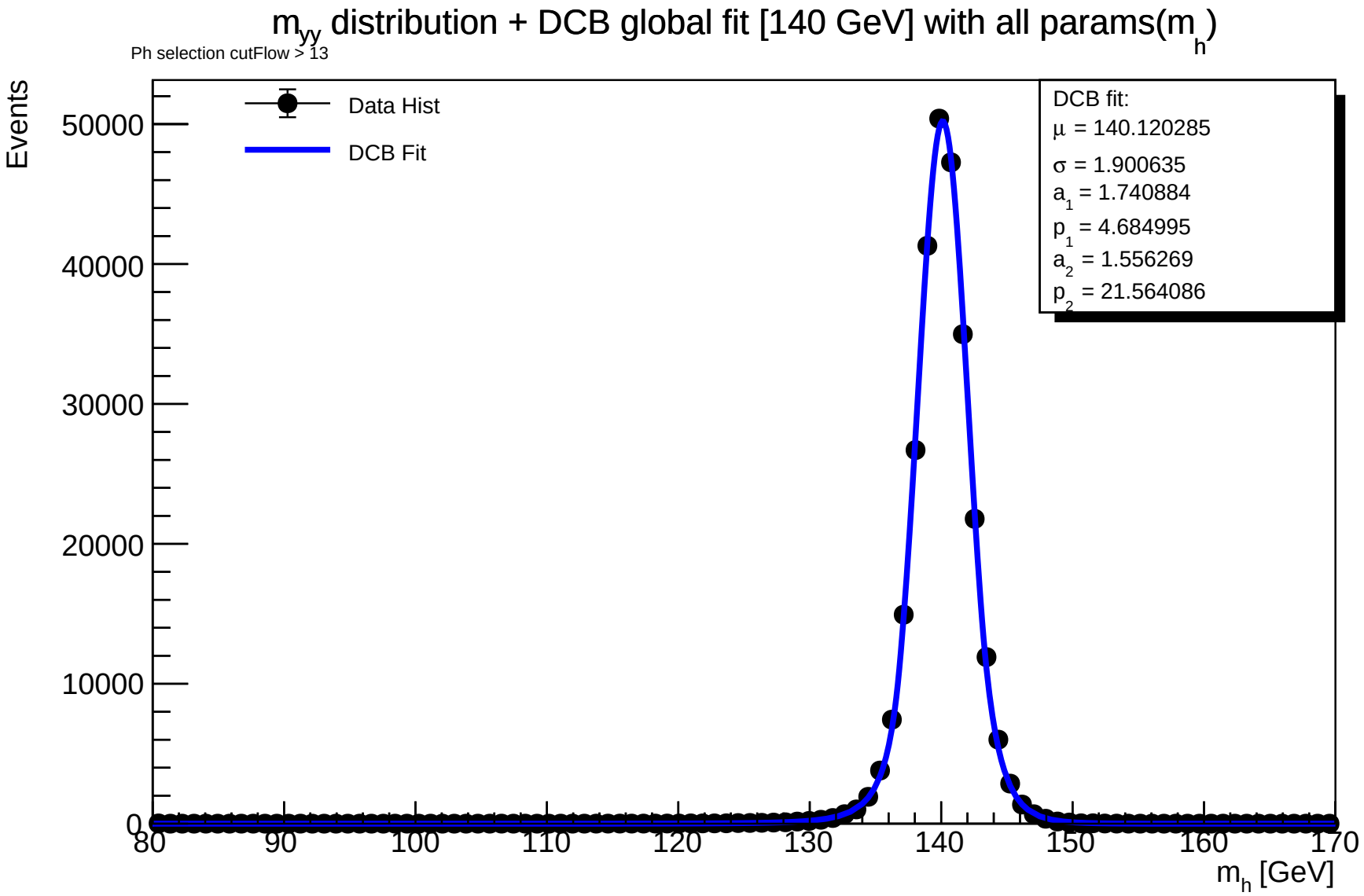
Global fit all params(m_H) [125 GeV]



Global fit all params(m_H) [130 GeV]

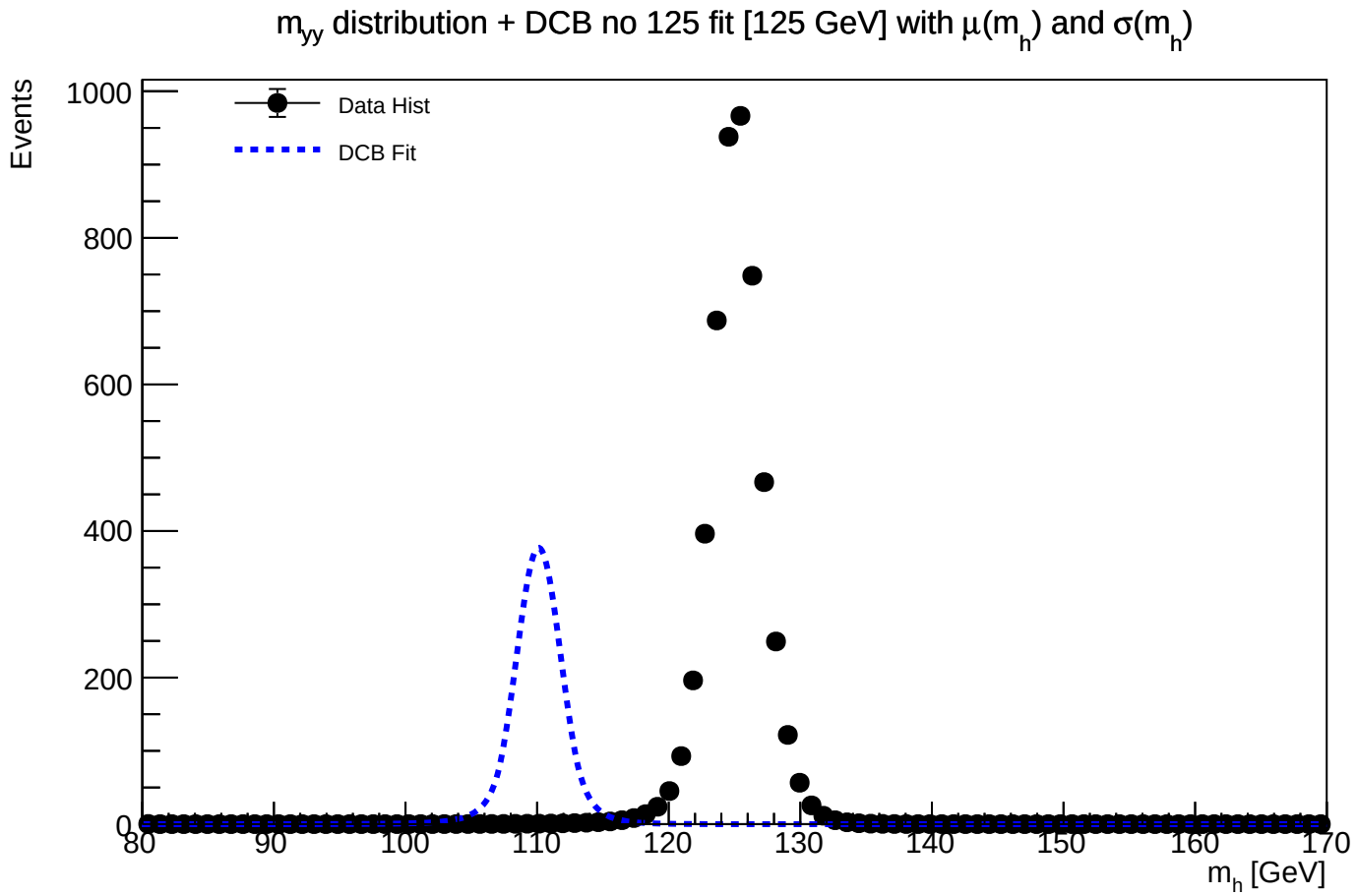


Global fit all params(m_H) [140 GeV]



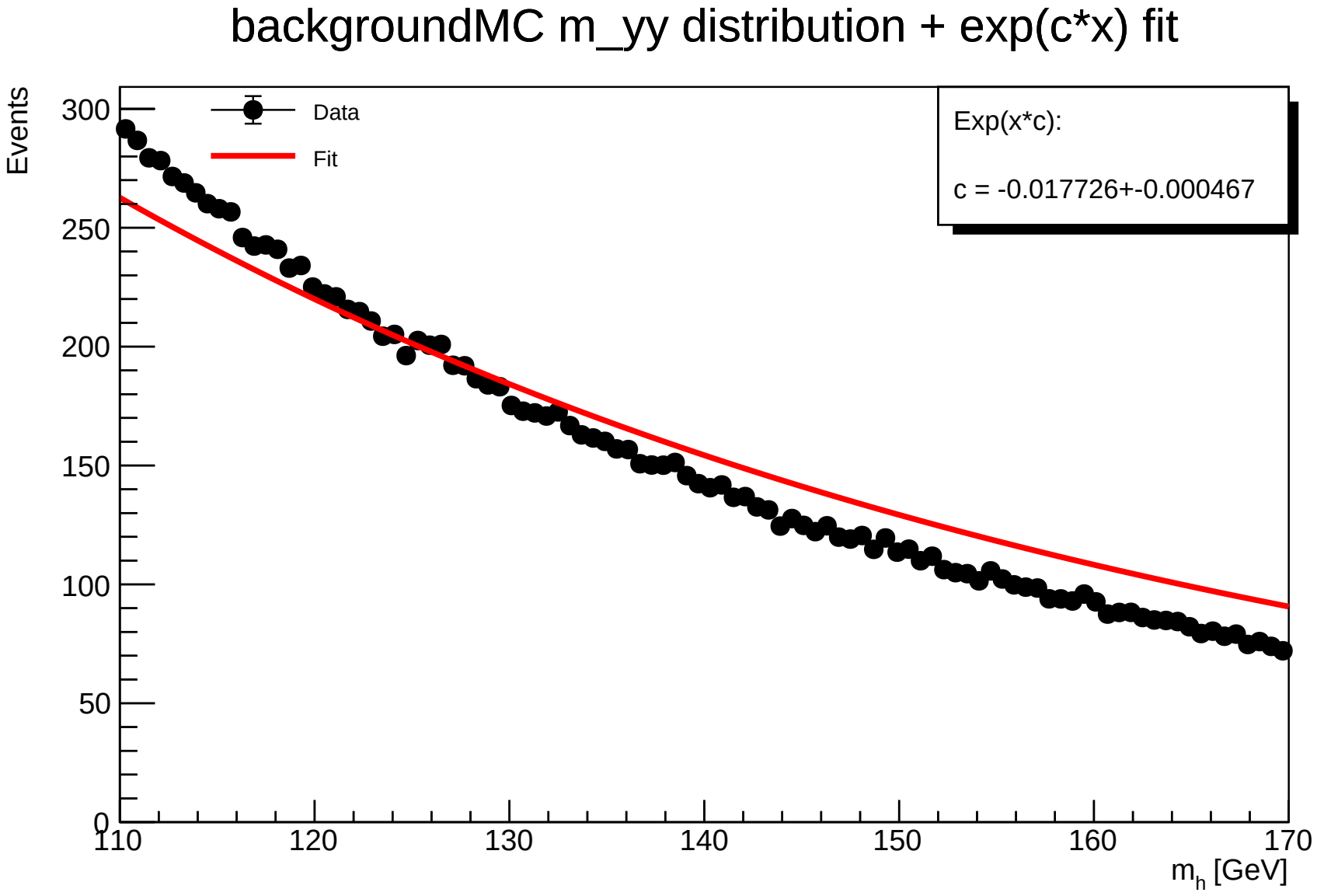
Fit without 125 GeV dataset

- I have created a fit using 110, 130, 140 GeV datasets;
- Problem: How to apply this fit to the 125 GeV dataset?:
  - create a new fit with the parameters fixed by the previous fit;



Add error bars

I have read the documentation at this link:

- there is the function `getPropagatedError(const RooFitResult fr)` to obtain the propagated errors
- I tried to create a `RooFitResult` after a `RooSimultaneous` fit, but it does not work;

Background MC